

SOC

*Power law and Self-Organized
Criticality*

Power laws, Pareto distributions and Zipf's law

M. E. J. Newman

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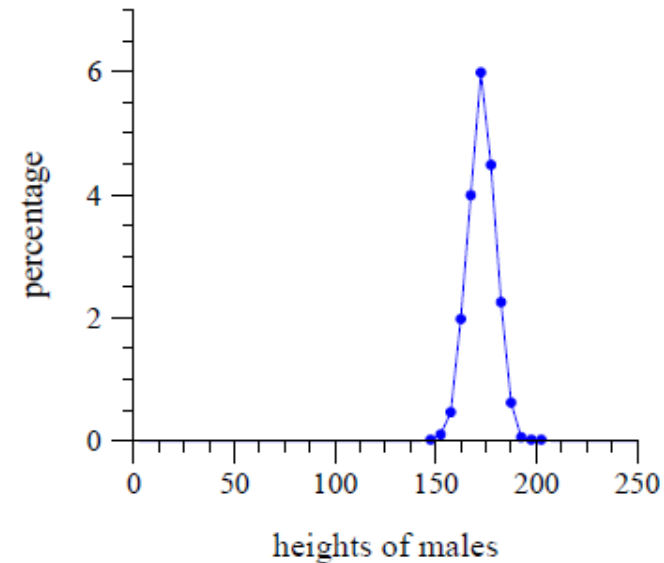
- *Many of the things that scientists measure **have a typical size or “scale”—a typical value** around which individual measurements are centred.*

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- *Many of the things that scientists measure **have a typical size or “scale”**—a typical value around which individual measurements are centred.*
- **Histogram** of people's heights



Heights in centimetres of American males. Data from the National Health Examination Survey, 1959–1962 (US Department of Health and Human Services).

<https://aaronclauset.github.io/powerlaws/data.htm>

Histogram : Linear binning

Key Steps in Linear Binning

- 1. Define the Range:** *Determine the minimum and maximum values of the dataset.*
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Suppose you have a dataset of values:

[2, 5, 7.3, 13, 15]

- **Range:** Minimum = 2, Maximum = 15
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Data points in bins:

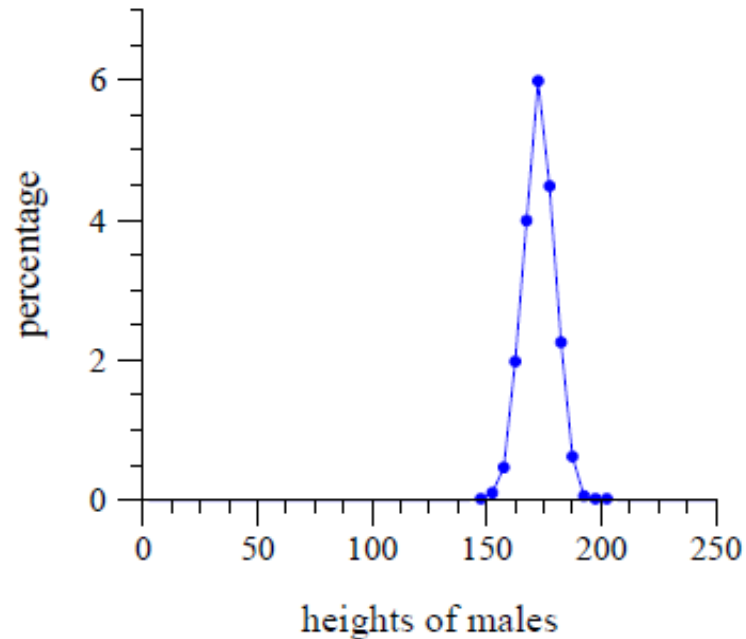
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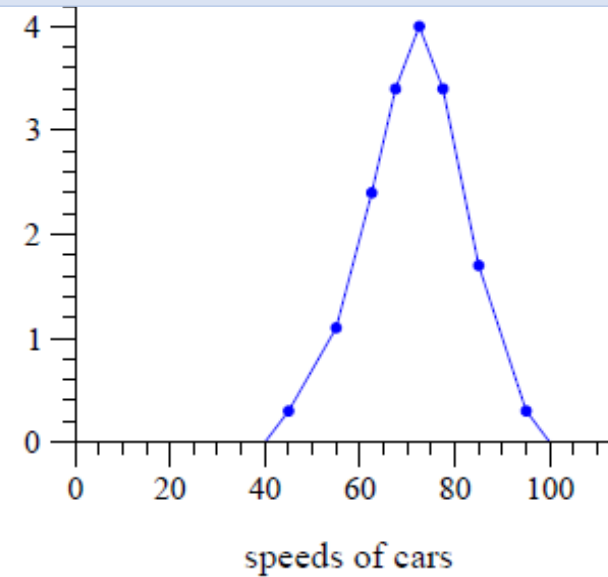
Applications

- **Histograms:** *To visualize the distribution of data.*
- **Data Compression:** *Reducing the number of data points by grouping them into bins.*
- **Signal Processing:** *Discretizing continuous signals.*
- **Numerical Methods:** *Approximating functions or integrals.*

- Many of the things that scientists measure *have a typical size or “scale”—a typical value* around which individual measurements are centred.



Again the histogram of speeds is strongly peaked, in this case around 75mph.



Right: histogram of speeds in miles per hour of cars on UK motorways. Data from Transport Statistics 2003 (UK Department for Transport).

Power laws, Pareto distributions and Zipf's law

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But *not all things* we measure *are peaked around a typical value*. Some vary over an enormous dynamic range, sometimes *many orders of magnitude!!!!*

- **frequency table** of words in a text.

$$\text{word frequency} \propto \frac{1}{\text{word rank}} .$$

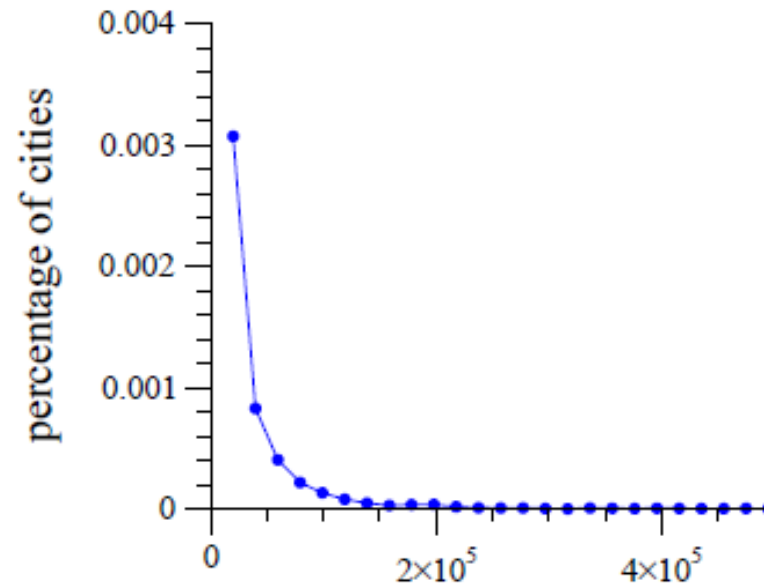


Zipf's law (/zɪf/; German pronunciation: [tsɪpf]) is an *empirical law* stating that when a list of measured values is sorted in decreasing order, the value of the n -th entry is often approximately *inversely proportional* to n .

George Kingsley Zipf (Statistics, linguistics)

*But **not all things** we measure **are peaked around a typical value**. Some vary over an enormous dynamic range, sometimes **many orders of magnitude**!!!!*

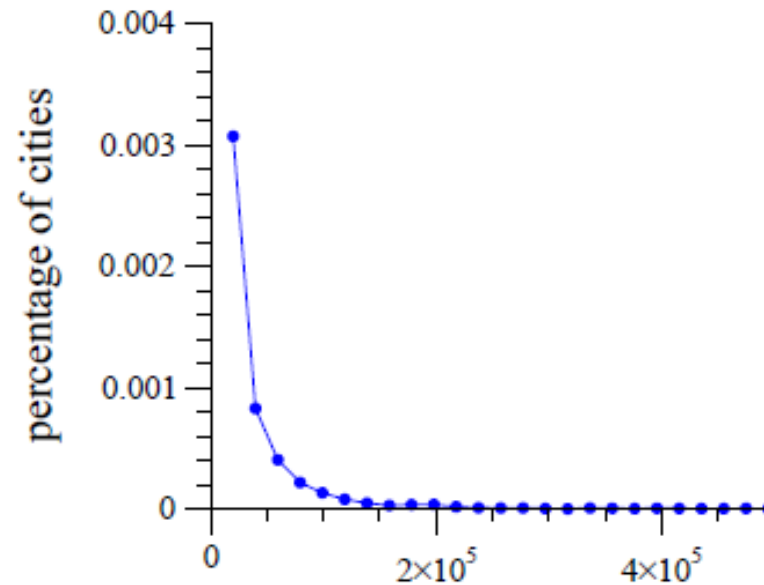
But *not all things* we measure *are peaked around a typical value*. Some vary over an enormous dynamic range, sometimes *many orders of magnitude!!!!*



Histogram of the populations of all US cities with population of 10 000 or more.

The largest city in the US is New York City, with over 8.5 million residents. Los Angeles and Chicago follow, each with more than 2.5 million residents, and southern US cities Houston and Phoenix round out the top five with populations of almost 2.3 million and 1.6 million, respectively.

But *not all things* we measure *are peaked around a typical value*. Some vary over an enormous dynamic range, sometimes *many orders of magnitude!!!!*

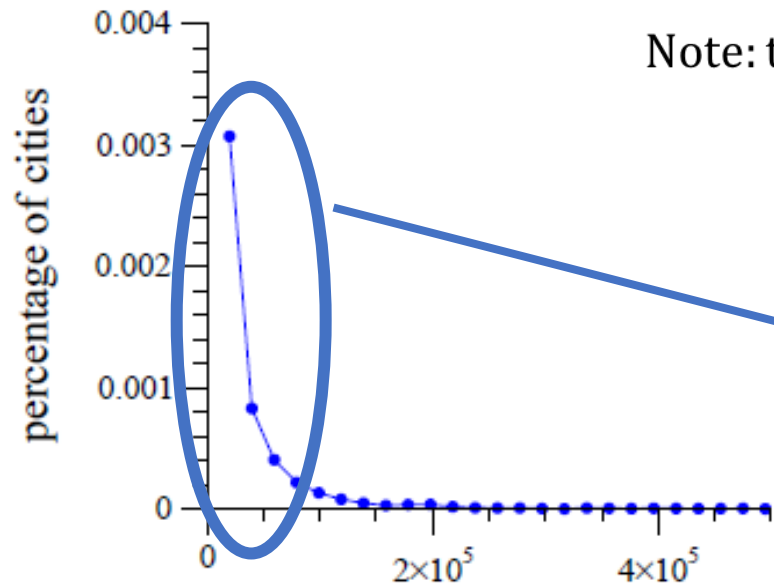


Histogram of the populations of all US cities with population of 10 000 or more.

Monowi,
Nebraska

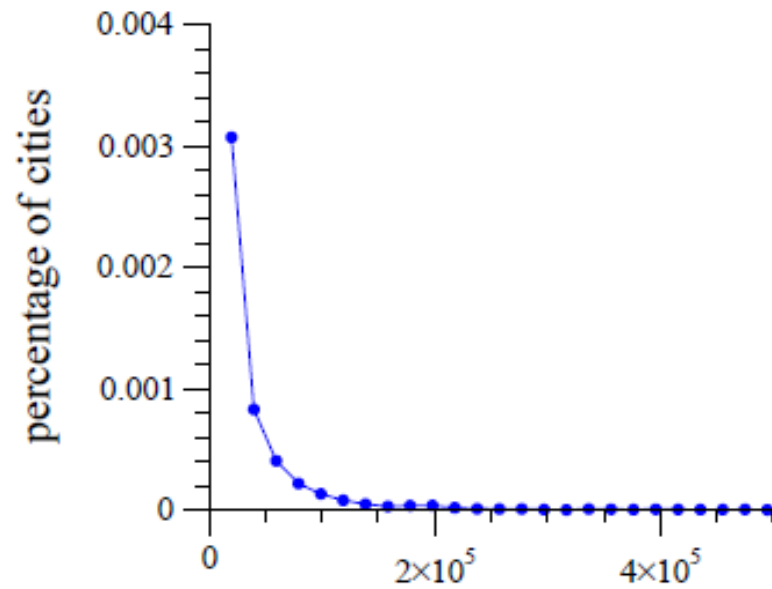
During the 2000 census, the village had a total population of two; only one married couple, Rudy and Elsie Eiler, lived there. Rudy died in 2004, leaving his wife as the only resident in the village. In this capacity, she acts as mayor, and granted herself a liquor license. She is required to produce a municipal road plan every year in order to secure state funding for the village's four street lights.

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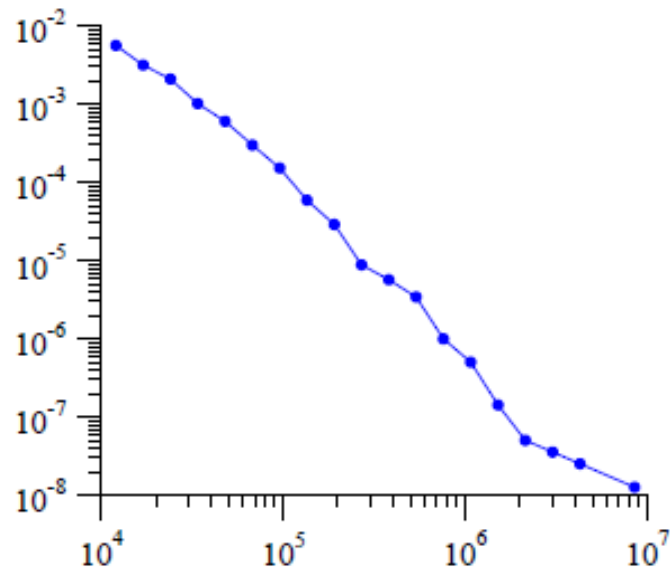


Note: the ratio of largest to smallest population is at least $\sim 10^5$.

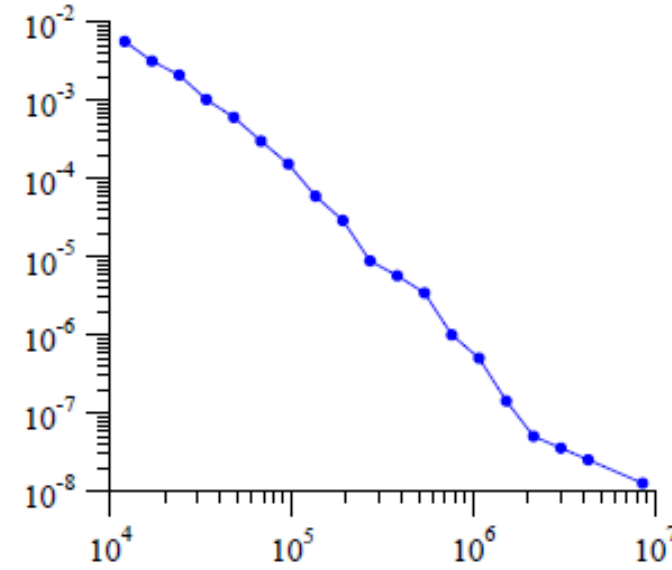
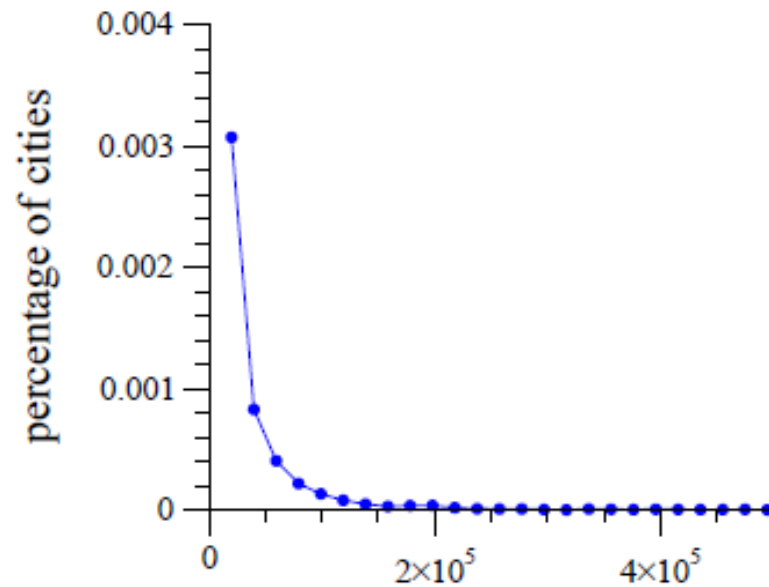
The histogram is *highly right-skewed*, meaning that while the *bulk of the distribution occurs for fairly small sizes* most US cities have small populations—there is a small number of cities with population much higher than the typical value, producing the *long tail to the right* of the histogram.



This right-skewed form is qualitatively quite different from the histograms of people's heights, but is not itself very surprising. Given that we know there is a large dynamic range from the smallest to the largest city sizes, we can immediately deduce that there can only be a small number of very large cities



*The histogram of city sizes again, but this time replotted with **logarithmic horizontal and vertical axes**. Now a remarkable pattern emerges: the histogram, when plotted in this fashion, **follows quite closely a straight line**.*

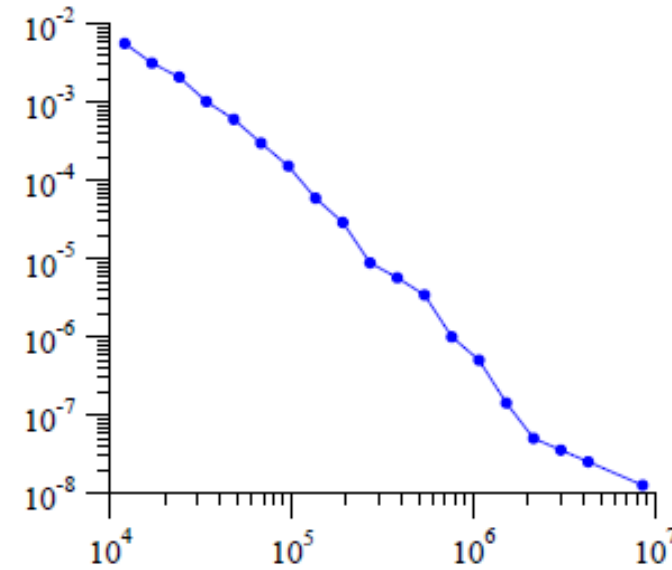
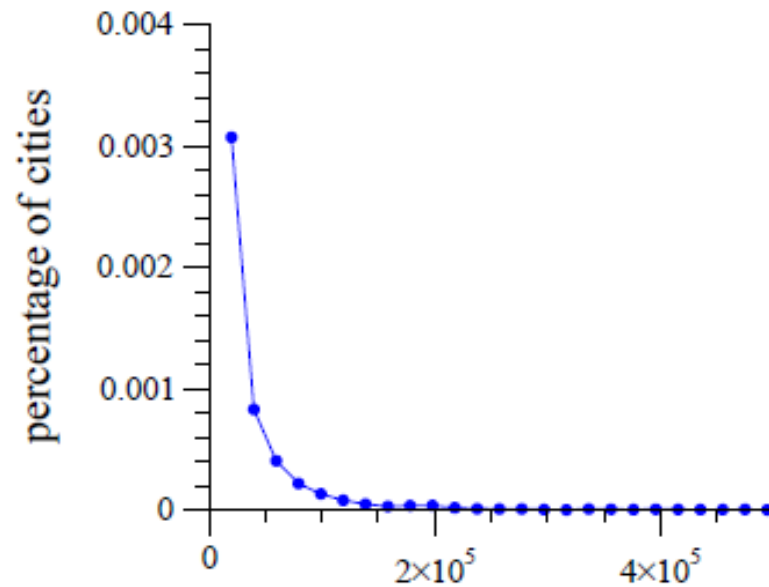


This observation seems first to have been made by Auerbach, although it is often attributed to [Zipf](#). What does it mean?

Let $p(x)dx$ be the fraction of cities with population between x and $x + dx$.

If the histogram is a straight line on log-log scale, then $\ln p(x) = -\alpha \ln(x) + c$, where α and c are constants.

- The slope is negative here.
- This is actually $p(x) = C x^{-\alpha}$, where $c = e^c$,
- This follows a power law.
- (sometimes called as scale-free). Why?



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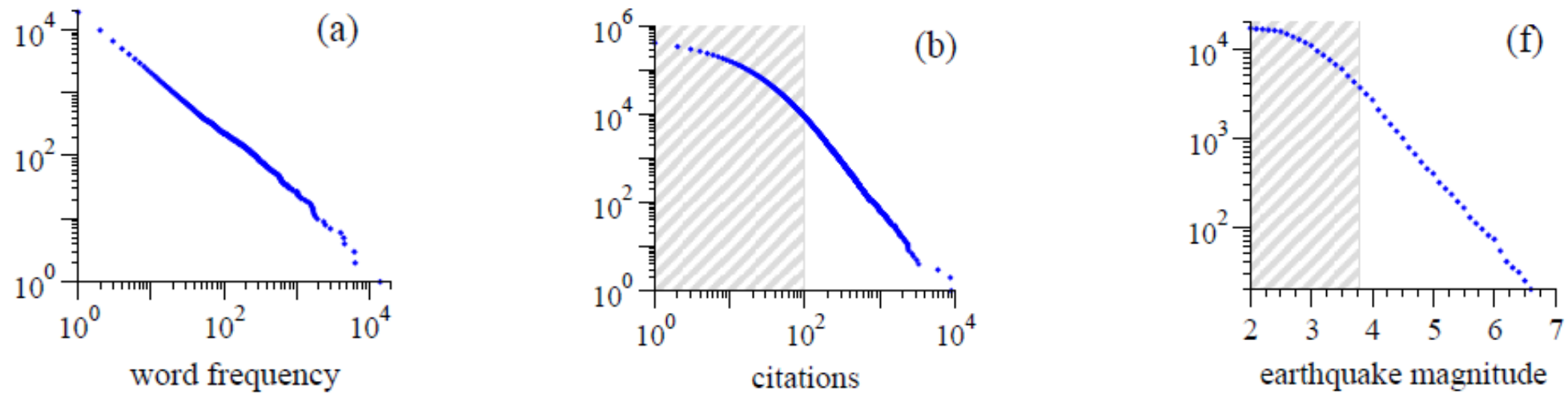
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Check it with all Indian states.

Track the population of each district in your states, and check whether the histogram is following power-law



Cumulative distributions or “rank/frequency plots”

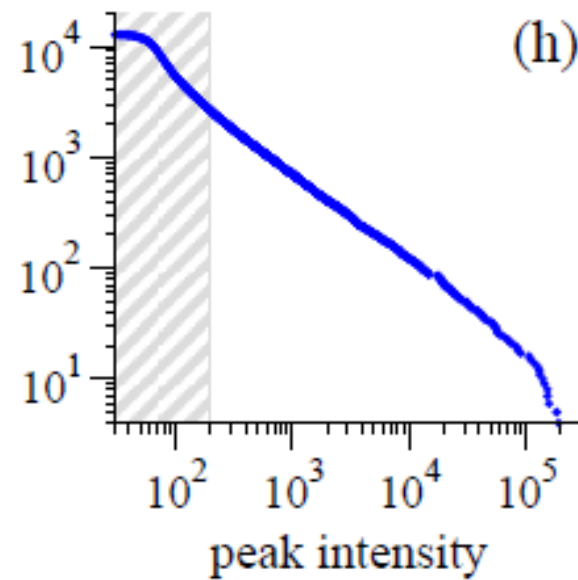
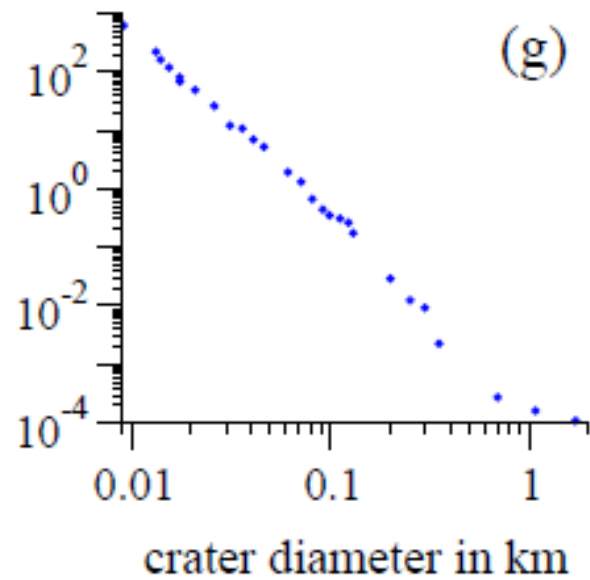
(a) Numbers of occurrences of unique words in the novel Moby Dick by Hermann Melville.

(b) Numbers of citations to scientific papers published in 1981, from time of publication until June 1997.

(f) Magnitude of earthquakes in California between January 1910 and May 1992. The Richter magnitude is defined as the logarithm, base 10, of the maximum amplitude of motion detected in the earthquake, and hence the horizontal scale in the plot, which is drawn as linear, is in effect a logarithmic scale of amplitude. The power law relationship in the earthquake distribution is thus a relationship between amplitude and frequency of occurrence. The data are from the National Geophysical Data Center, www.ngdc.noaa.gov.

<https://aaronclauset.github.io/powerlaws/data.html>

<https://physics.bu.edu/~redner/projects/citation/isi.html>



quantity	minimum	exponent
	$x_{\min}$	α
(a) frequency of use of words	1	2.20(1)
(b) number of citations to papers	100	3.04(2)
(c) number of hits on web sites	1	2.40(1)
(d) copies of books sold in the US	2 000 000	3.51(16)
(e) telephone calls received	10	2.22(1)
(f) magnitude of earthquakes	3.8	3.04(4)
(g) diameter of moon craters	0.01	3.14(5)
(h) intensity of solar flares	200	1.83(2)
(i) intensity of wars	3	1.80(9)
(j) net worth of Americans	\$600m	2.09(4)
(k) frequency of family names	10 000	1.94(1)
(l) population of US cities	40 000	2.30(5)

(g) Diameter of craters on the moon. Vertical axis is measured per square kilometre.

(h) Peak gamma-ray intensity of solar flares in counts per second, measured from Earth orbit between February 1980 and November 1989.

The observations were made between 1980 and 1989 by the instrument known as the Hard X-Ray Burst Spectrometer aboard the Solar Maximum Mission satellite launched in 1980. The spectrometer used a CsI scintillation detector to measure gamma-rays from solar flares and the horizontal axis in the figure is calibrated in terms of scintillation counts per second from this detector. The data are from the NASA Goddard Space Flight Center, umbra.nascom.nasa.gov/smm/hxrbs.html.

Indian population distribution (district wise)

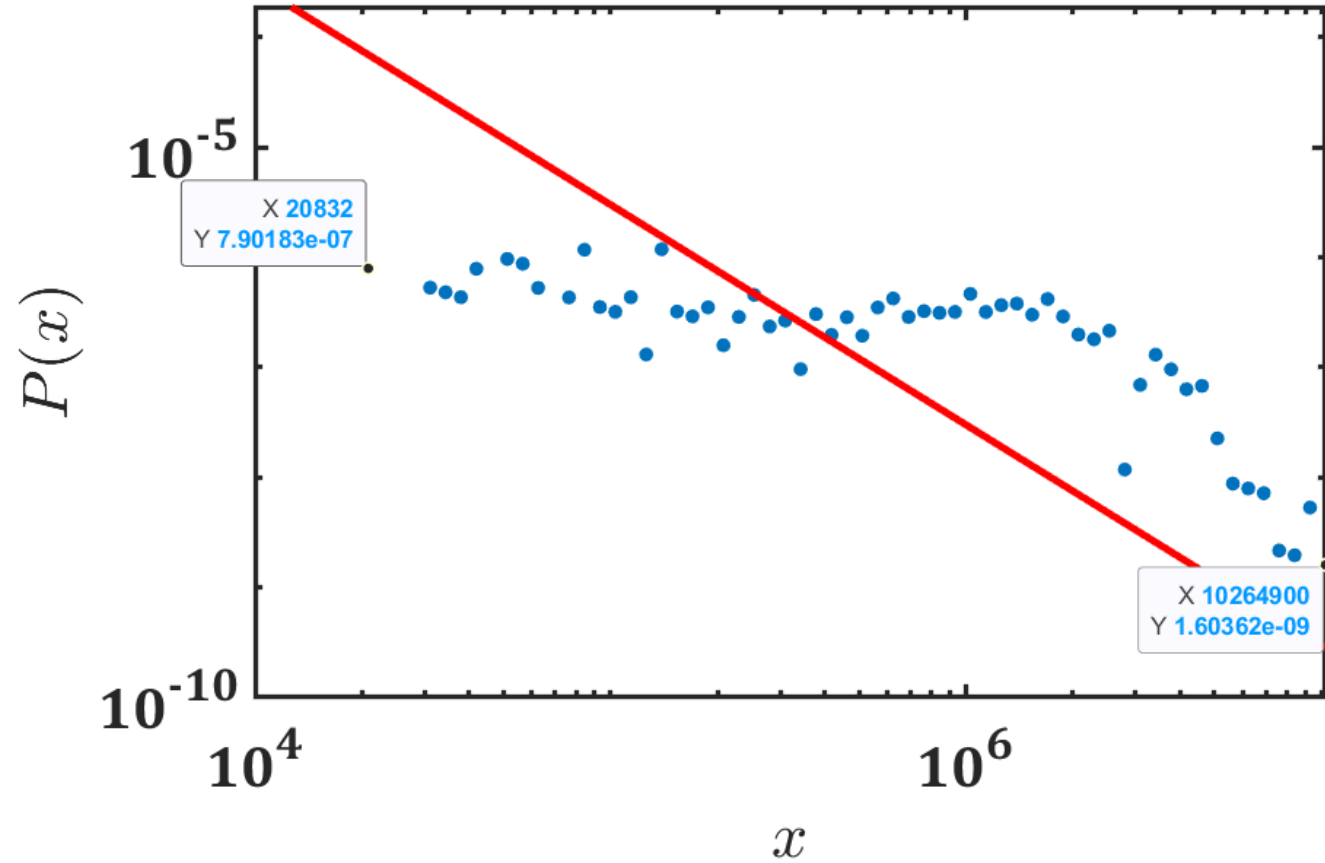
Urban populations → close to Zipf's law

- District populations → often lognormal with a heavy tail

- Megadistricts (Thane, North 24 Parganas, etc.) → power-law tail

- So the overall structure is typically:

Lognormal core + Power-law tail



Power-law distributions occur in an extraordinarily diverse range of phenomena. In addition to city populations,

- *The sizes of earthquakes [3],*
- *Moon craters [4],*
- *Solar flares [5],*
- *Computer files [6] and wars [7],*
- *The frequency of use of words in any human language [2, 8],*
- *The frequency of occurrence of personal names in most cultures [9], the numbers of papers scientists write [10], the number of citations received by papers [11],*
- *The number of hits on web pages [12],*
- *The sales of books, music recordings and almost every other branded commodity [13, 14],*
- *The numbers of species in biological taxa [15],*
- *People's annual incomes [16] follows power law distributions.*

[3] B. Gutenberg and R. F. Richter, *Frequency of earthquakes in California*. *Bulletin of the Seismological Society of America* 34, 185–188 (1944).

[4] G. Neukum and B. A. Ivanov, *Crater size distributions and impact probabilities on Earth from lunar, terrestrial planet, and asteroid cratering data*. In T. Gehrels (ed.), *Hazards Due to Comets and Asteroids*, pp. 359–416, University of Arizona Press, Tucson, AZ (1994).

[5] E. T. Lu and R. J. Hamilton, *Avalanches of the distribution of solar flares*. *Astrophysical Journal* 380, 89–92 (1991).

[6] M. E. Crovella and A. Bestavros, *Self-similarity in World Wide Web traffic: Evidence and possible causes*. In B. E. Gaither and D. A. Reed (eds.), *Proceedings of the 1996 ACM SIGMETRICS Conference on Measurement and Modeling of Computer Systems*, pp. 148–159, Association of Computing Machinery, New York (1996).

[7] D. C. Roberts and D. L. Turcotte, *Fractality and self-organized criticality of wars*. *Fractals* 6, 351–357 (1998).

[8] J. B. Estoup, *Gammes Sténographiques*. Institut Sténographique de France, Paris (1916).

[9] D. H. Zanette and S. C. Manrubia, *Vertical transmission of culture and the distribution of family names*. *Physica A* 295, 1–8 (2001).

[10] A. J. Lotka, *The frequency distribution of scientific production*. *J. Wash. Acad. Sci.* 16, 317–323 (1926).

[11] D. J. de S. Price, *Networks of scientific papers*. *Science* 149, 510–515 (1965).

[12] L. A. Adamic and B. A. Huberman, *The nature of markets in the World Wide Web*. *Quarterly Journal of Electronic Commerce* 1, 512 (2000).

[13] R. A. K. Cox, J. M. Felton, and K. C. Chung, *The concentration of commercial success in popular music: an analysis of the distribution of gold records*. *Journal of Cultural Economics* 19, 333–340 (1995).

[14] R. Kohli and R. Sah, *Market shares: Some power law results and observations*. Working paper 04.01, Harris School of Public Policy, University of Chicago (2003).

[15] J. C. Willis and G. U. Yule, *Some statistics of evolution and geographical distribution in plants and animals, and their significance*. *Nature* 109, 177–179 (1922).

[16] V. Pareto, *Cours d'Economie Politique*. Droz, Geneva (1896).

Histogram : Linear binning

Key Steps in Linear Binning

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Suppose you have a dataset of values:

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Histogram : Log binning

- *Log binning is a technique used in data analysis, particularly for dealing with data that **spans several orders of magnitude**, such as distributions that follow **power laws**.*
- *Instead of using linear bin widths, the **bins are spaced logarithmically**, allowing better representation of data in cases where the frequency of occurrence decreases exponentially with increasing values.*

Histogram : Log binning

Key Features of Log Binning:

1. Logarithmic Bin Widths:

1. The width of the bins *increases logarithmically*, meaning the difference between bin boundaries grows as the values increase.
2. For example, bin edges might be spaced like 1, 10, 100, 1000 rather than 1, 2, 3, 4.

2. Preserves Detail Across Scales:

1. Small bins are used for low-value ranges, capturing fine details.
2. Larger bins are used for high-value ranges, ensuring sparsely populated regions are still well-represented.

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3. Reduces Noise:

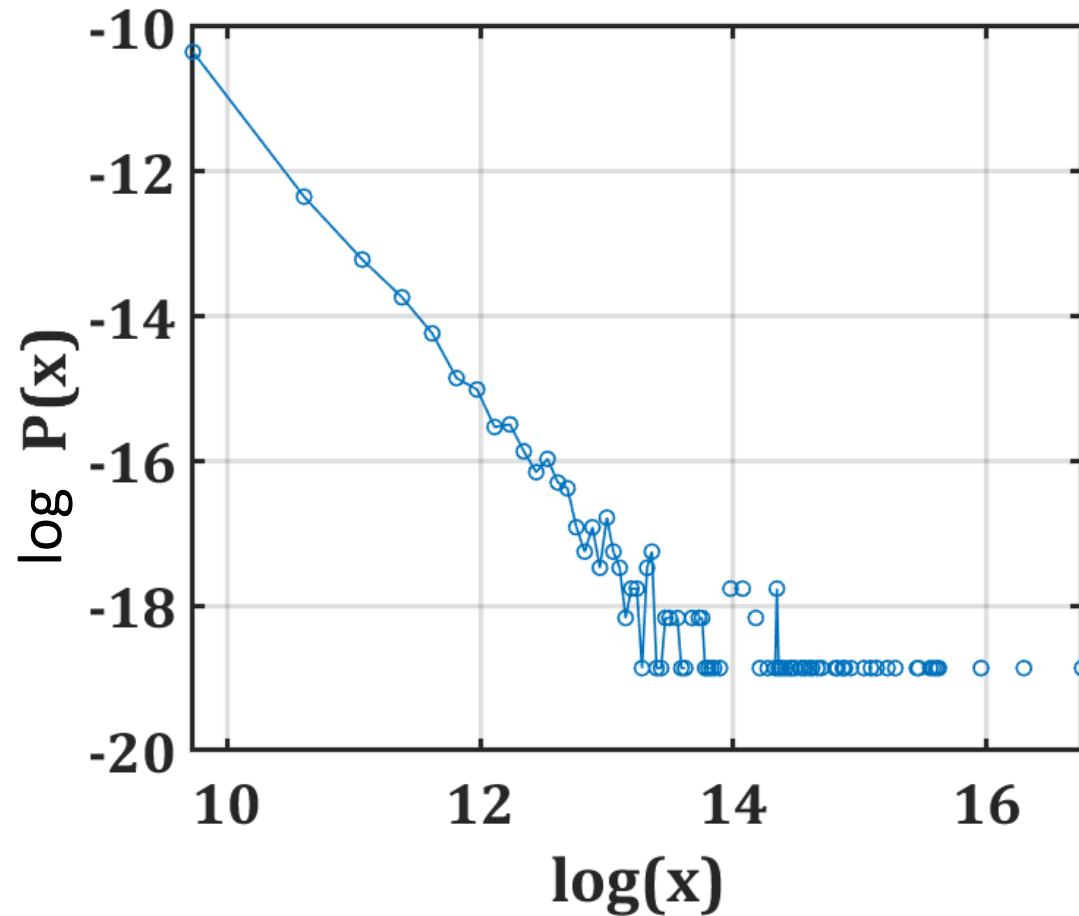
1. In distributions with a long tail (e.g., power-law distributions), linear binning can lead to sparse or empty bins for large values. Log binning aggregates these values more effectively, reducing statistical noise.

4. Logarithmic Representation:

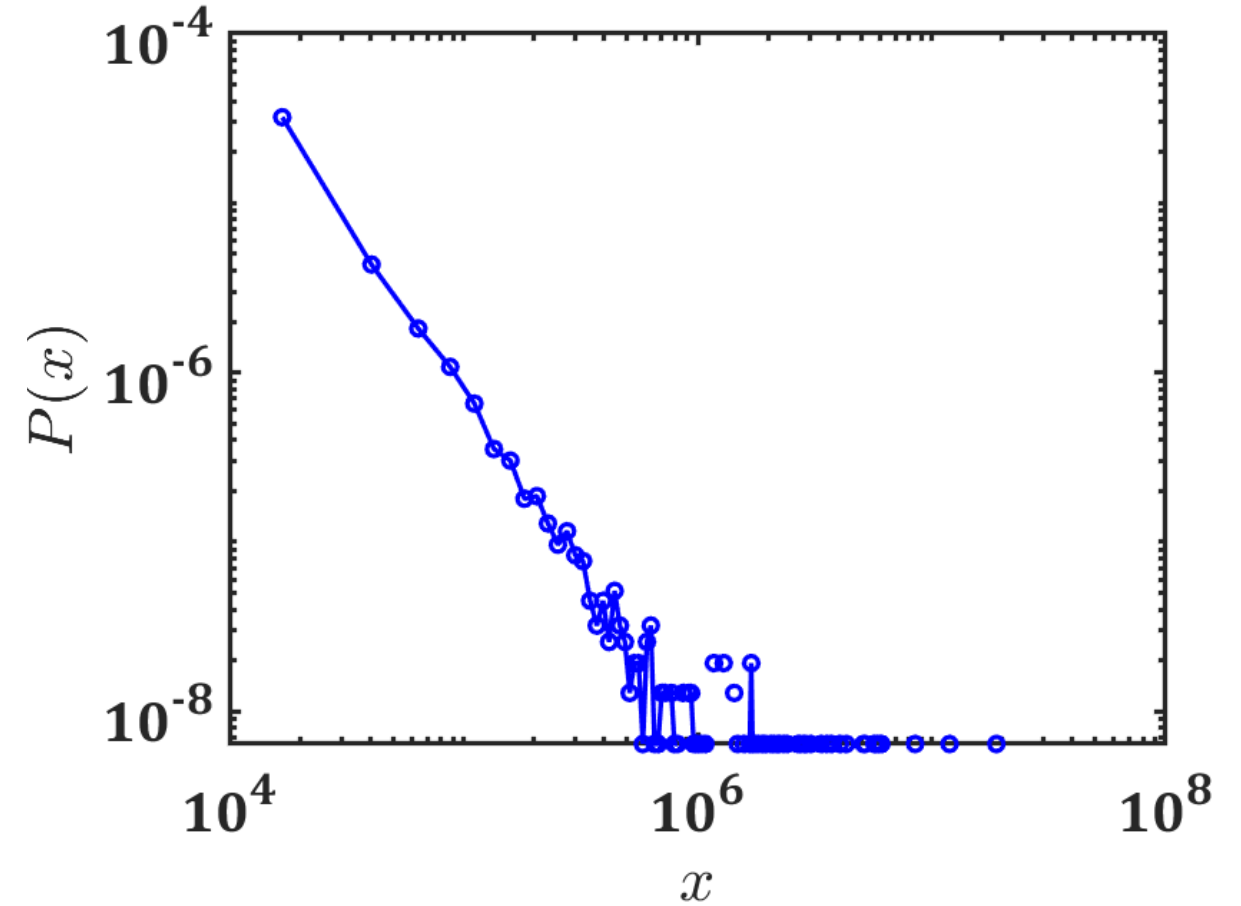
1. Data is often plotted on a log-log scale for visualization. The x-axis represents the bin range, and the y-axis shows the frequency or probability density.

Histogram (US population)

Linear binning and log of the binned data

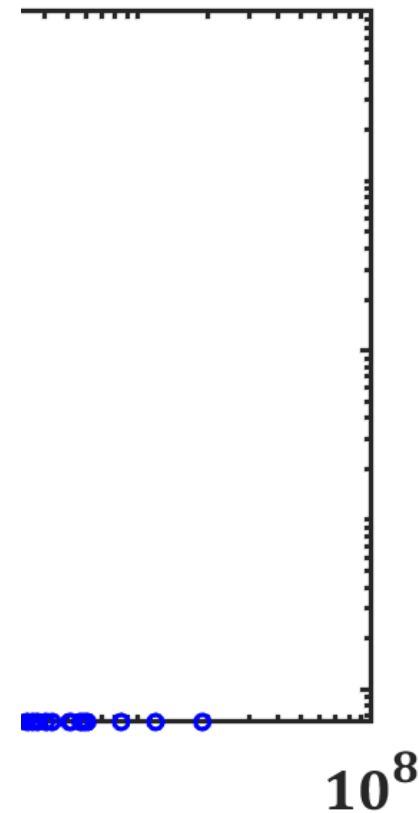
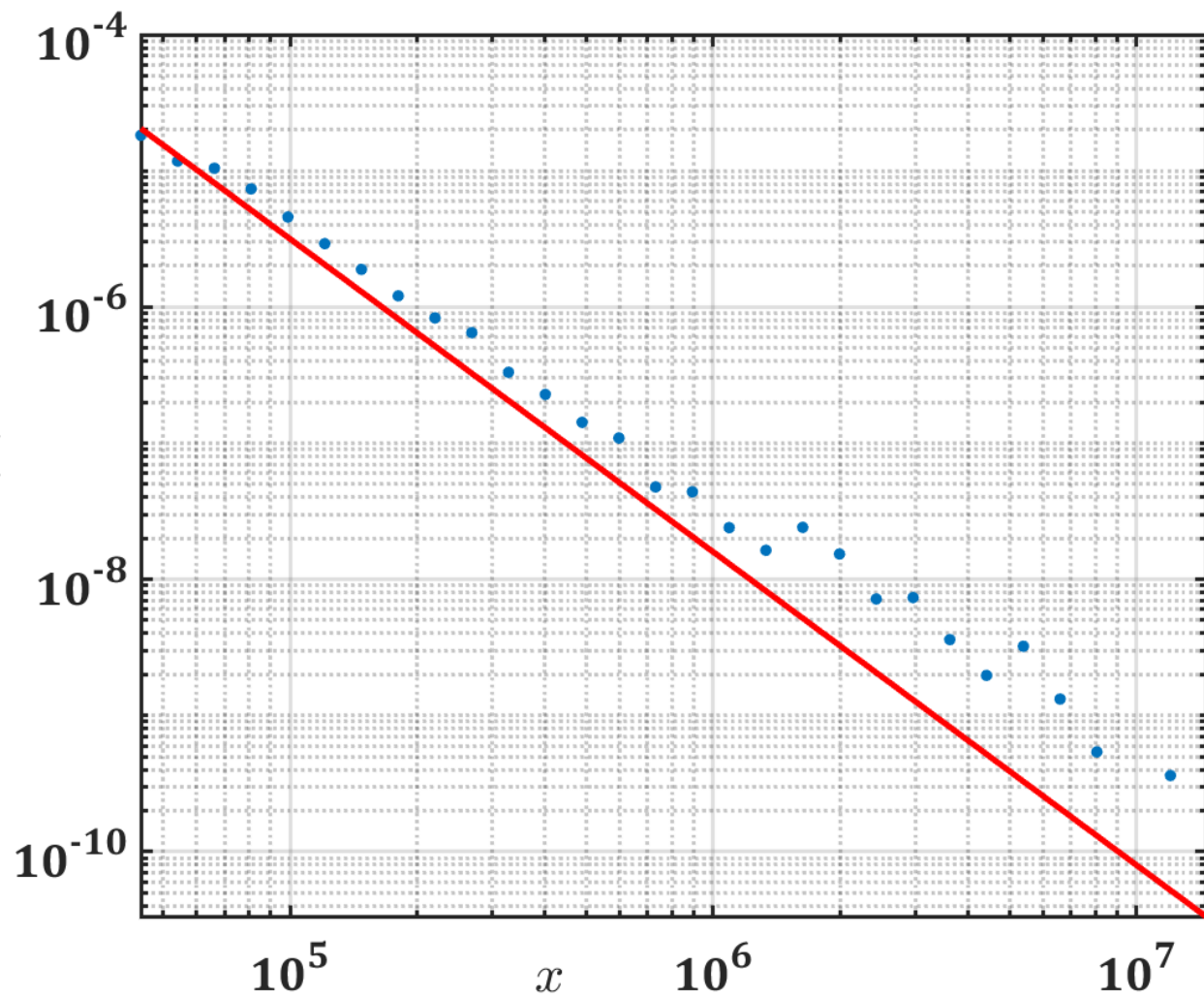
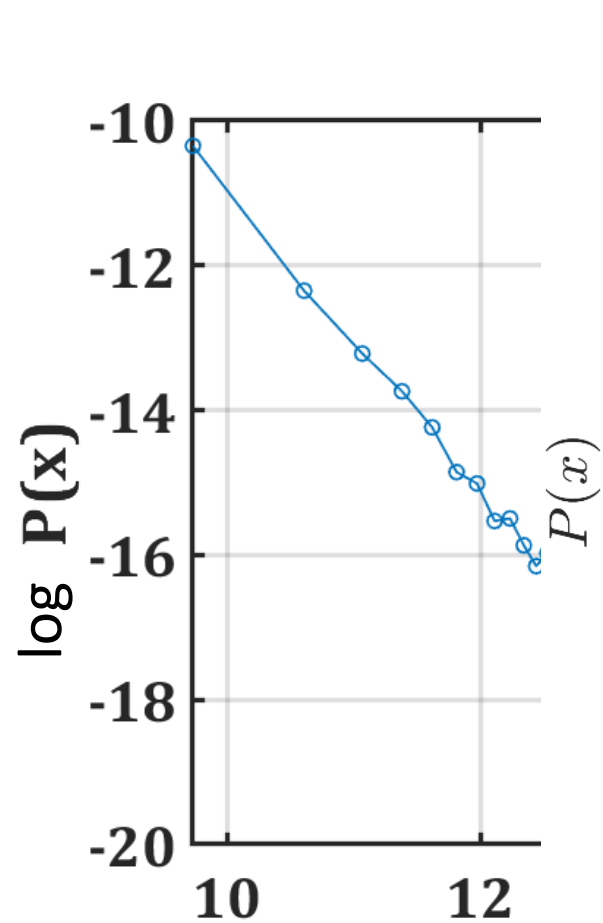


Linear binning and log-scale



Histogram (US population)

Log binning and log scale



Linear vs Logarithmic Binning

Feature	Linear Binning			Logarithmic Binning		
Bin spacing	Feature Bin spacing Bin width Raw histogram count Normalization step 1 Normalization step 2 Final quantity plotted Reason Typical use	Feature Bin spacing Bin width Raw histogram count Normalization step 1 Normalization step 2 Final quantity plotted Reason Typical use	Feature Bin spacing Bin width Raw histogram count Normalization step 1 Normalization step 2 Final quantity plotted Reason Typical use	Feature Bin spacing Bin width Raw histogram count Normalization step 1 Normalization step 2 Final quantity plotted Reason Typical use	Feature Bin spacing Bin width Raw histogram count Normalization step 1 Normalization step 2 Final quantity plotted Reason Typical use	Feature Bin spacing Bin width Raw histogram count Normalization step 1 Normalization step 2 Final quantity plotted Reason Typical use
Bin width	Constant					
Raw histogram count						
Normalization step 1						
Normalization step 2						
Final quantity plotted						
Reason	Convert counts $\rightarrow$ probability density			Correct for different bin sizes		
Typical use	Gaussian-like distributions			Power-law / heavy-tail distributions		
Feature	Linear Binning	Logarithmic Binning	Logarithmic Binning	Feature	Linear Binning	Logarithmic Binning
Bin spacing	Equal spacing in x	Equal spacing in $\log(x)$	Increasing with x	Bin spacing	Equal spacing in x	Equal spacing in $\log(x)$
Bin width	Constant	Increasing with x	Constant	Bin width	Constant	Increasing with x
Raw histogram count	n_i	n_i	n_i	Raw histogram count	n_i	n_i
Normalization step 1	Divide by total samples N	Divide by total samples N	Divide by total samples N	Normalization step 1	Divide by total samples N	Divide by total samples N
Normalization step 2	Divide by bin width Δx	Divide by bin width Δx , each bin different	Divide by bin width Δx , each bin different	Normalization step 2	Divide by bin width Δx	Divide by bin width Δx , each bin different
Final quantity plotted	$\frac{n_i}{N\Delta x}$	$\frac{n_i}{N\Delta x_i}$	$\frac{n_i}{N\Delta x_i}$	Final quantity plotted	$\frac{n_i}{N\Delta x}$	$\frac{n_i}{N\Delta x_i}$
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Histogram : Log binning

- 1. Choose the Base of the Logarithm:** Decide the base for your logarithmic binning (e.g., base-2, base-10, or natural log). This determines the width of the bins.
- 2. Determine the Range of Data:** Identify the minimum and maximum values of the dataset. These will define the range for your bins.
- 3. Define the Bin Edges:** Create bin edges using logarithmic intervals. For instance, if the base is 10:
Bin edges = $\{10^0, 10^1, 10^2, \dots\}$
Or, for smaller steps:
Bin edges = $\{10^0, 10^{0.1}, 10^{0.2}, \dots\}$
- 4. Assign Data Points to Bins:** For each data point, determine which bin it belongs to by comparing its value to the bin edges. In practice, you can use algorithms for efficient bin allocation.
- 5. Count Data Points in Each Bin:** After assigning all data points to bins, count the number of points in each bin.
- 6. Normalize Bin Counts :** Normalize the bin counts by the bin width to account for the fact that the bin sizes grow logarithmically.
- 7. Plot the Results:** Plot the histogram, often using a log-log scale to reveal power-law behavior or other patterns.

Histogram : Linear binning and Log binning

Task : Histogram of the set of random numbers which will have a power-law distribution with exponent $\alpha = 2.5$.

$$p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

Use transformation method.

Generate a random real number r uniformly distributed in the range $0 \leq r < 1$, then $x = x_{\min}(1 - r)^{-1/(\alpha-1)}$ is a random power-law-distributed real number in the range $x_{\min} \leq x < \infty$ with exponent α .

Note that there has to be a lower limit $x_{\min}$ on the range; the power-law distribution diverges as $x \rightarrow 0$

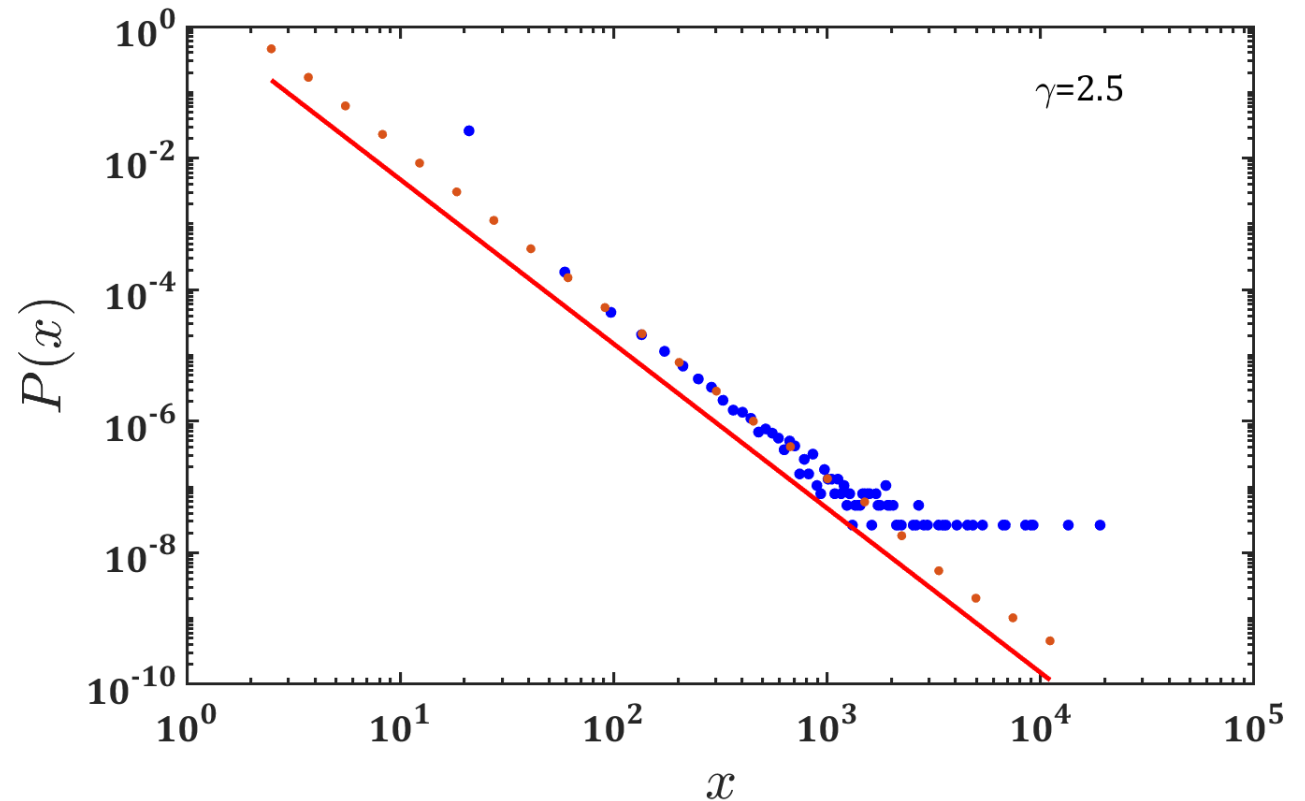
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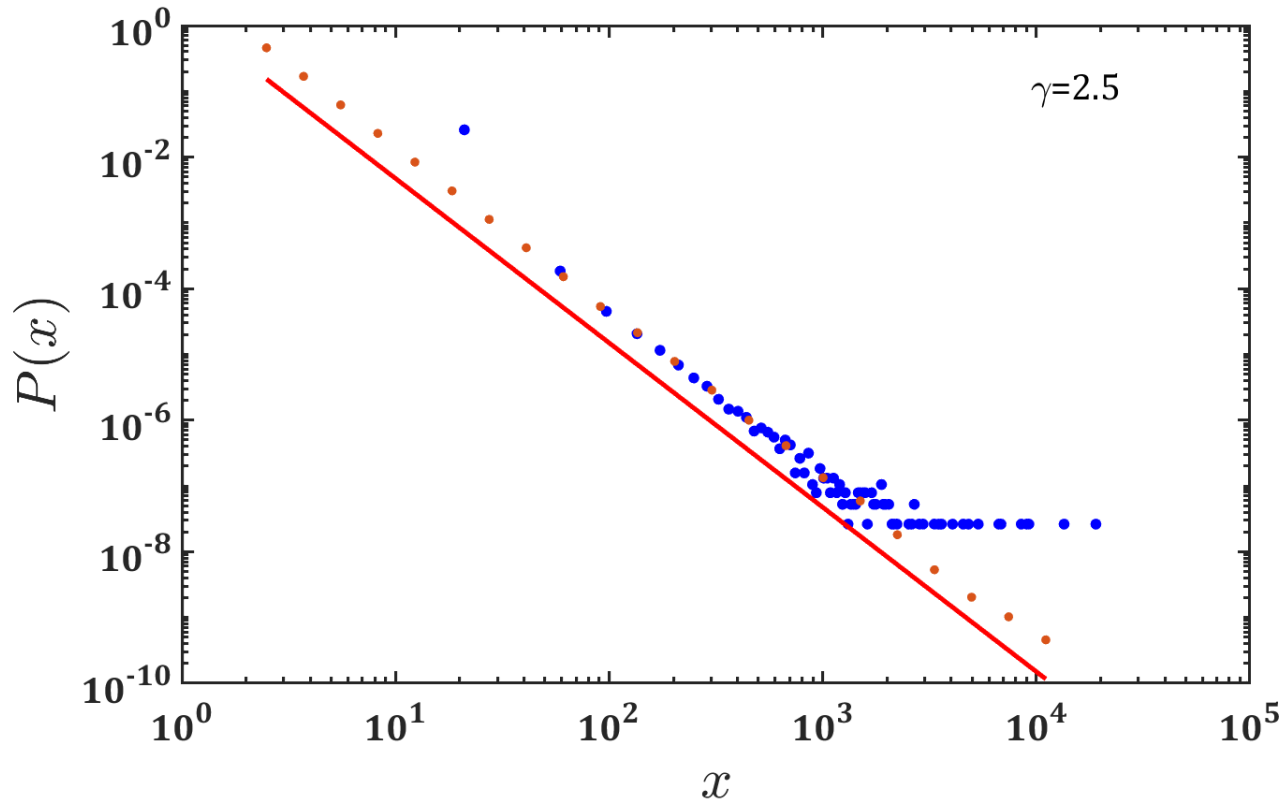
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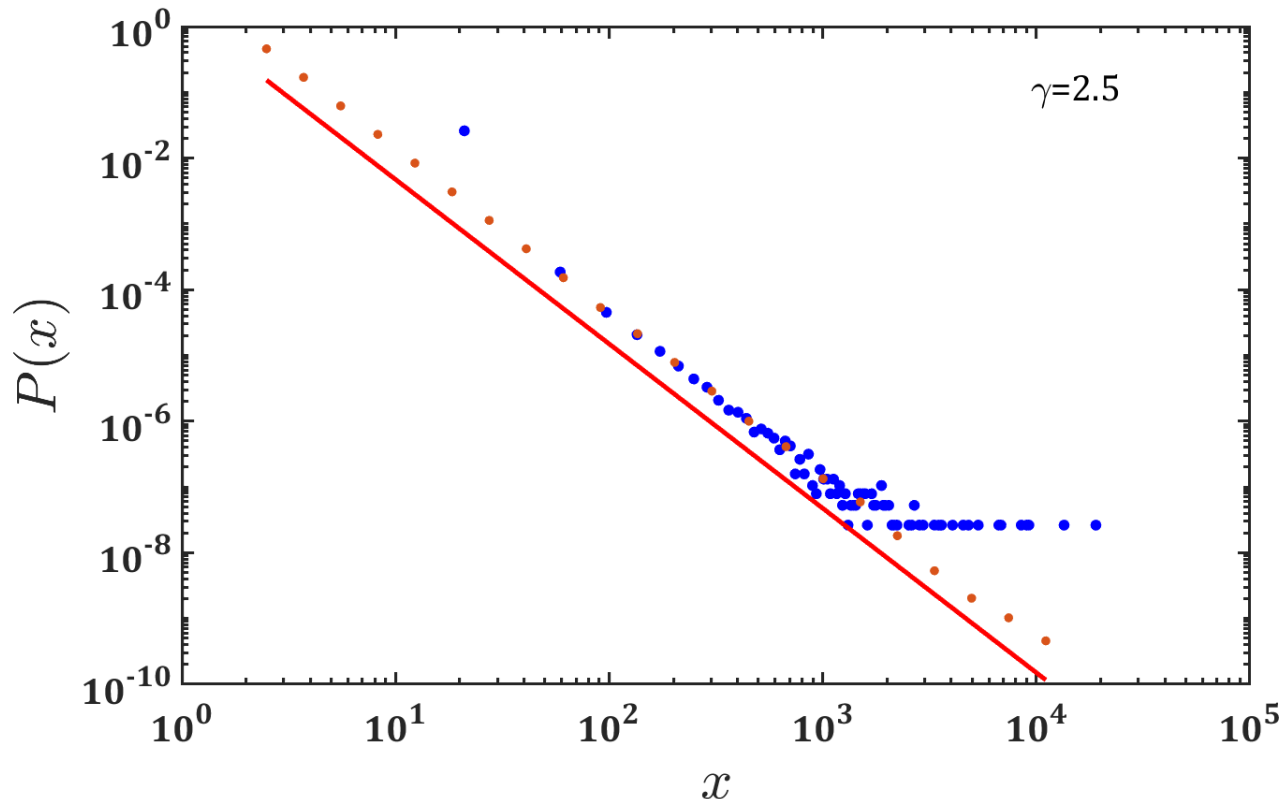
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- To reveal the power-law form of the distribution it is better, as we have seen, to plot the histogram on logarithmic scales, and when we do this for the current data we see the characteristic straight-line form of the power-law distribution (blue dots).
- However, the plot is in some respects not a very good one. In particular the right-hand end of the distribution is noisy because of sampling errors.
- The power-law distribution dwindles in this region, meaning that each bin only has a few samples in it, if any. So the fractional fluctuations in the bin counts are large and this appears as a noisy curve on the plot.

Histogram : Linear binning vs Log binning

Task : Histogram of the set of random numbers which will have a power-law distribution with exponent $\alpha = 2.5$.



- **Solution:** Vary the width of the bins in the histogram. If we are going to do this, we must also normalize the sample counts by the width of bins they fall in. **That is, the number of samples in a bin of width Δx should be divided by Δx to get a count per unit interval of x .**
- Then the normalized sample count **becomes independent of bin width on average** and we are free to vary the bin widths as we like.

The most common choice is to create bins such that each is a fixed multiple wider than the one before it. This is known as **logarithmic binning**.

A continuous real variable with a power-law distribution has a probability $p(x) \, \mathrm{d}x$ of taking a value in the interval from x to $x + \mathrm{d}x$, where

$$p(x) = Cx^{-\alpha},$$

with $\alpha > 0$.

The constant C is given by the normalization requirement that

$$1 = \int_{x_{\min}}^{\infty} p(x) \, \mathrm{d}x = C \int_{x_{\min}}^{\infty} x^{-\alpha} \, \mathrm{d}x = \frac{C}{1-\alpha} \left[x^{-\alpha+1} \right]_{x_{\min}}^{\infty}. \quad C = (\alpha - 1)x_{\min}^{\alpha-1},$$

$$p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

$$p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

The mean value of our power-law distributed quantity x is given by

$$\begin{aligned} \langle x \rangle &= \int_{x_{\min}}^{\infty} xp(x) \, dx = C \int_{x_{\min}}^{\infty} x^{-\alpha+1} \, dx \\ &= \frac{C}{2 - \alpha} \left[x^{-\alpha+2} \right]_{x_{\min}}^{\infty}. \end{aligned}$$

Note that this expression becomes infinite if $\alpha \leq 2$. Power laws with such low values of α have no finite mean. The distributions of sizes of solar flares and wars in are examples of such power laws.

$$p(x) = \frac{\alpha - 1}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha}.$$

For $\alpha > 2$ however, the mean is perfectly well defined, with a value given by

$$\langle x \rangle = \frac{\alpha - 1}{\alpha - 2} x_{\min}.$$

We can also calculate higher moments of the distribution $p(x)$. For instance, the second moment, the mean square, is given by

$$\langle x^2 \rangle = \frac{C}{3 - \alpha} \left[x^{-\alpha+3} \right]_{x_{\min}}^{\infty}.$$

For $\alpha > 2$ however, the mean is perfectly well defined, with a value given by Eq. (11) of

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We can also calculate higher moments of the distribution $p(x)$. For instance, the second moment, the mean square, is given by

$$\langle x^2 \rangle = \frac{C}{3 - \alpha} \left[x^{-\alpha+3} \right]_{x_{\min}}^{\infty}.$$

If $\alpha > 3$, then the second moment is finite and well-defined, taking the value

$$\langle x^2 \rangle = \frac{\alpha - 1}{\alpha - 3} x_{\min}^2.$$

SOC

Self-Organized Criticality



PHYSICAL REVIEW LETTERS

VOLUME 59

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NUMBER 4

Self-Organized Criticality: An Explanation of $1/f$ Noise

Per Bak, Chao Tang, and Kurt Wiesenfeld

Physics Department, Brookhaven National Laboratory, Upton, New York 11973

(Received 13 March 1987)

We show that dynamical systems with spatial degrees of freedom naturally evolve into a self-organized critical point. Flicker noise, or $1/f$ noise, can be identified with the dynamics of the critical state. This picture also yields insight into the origin of fractal objects.

PACS numbers: 05.40.+j, 02.90.+p

Self-organized criticality

Per Bak, Chao Tang, and Kurt Wiesenfeld

Department of Physics, Brookhaven National Laboratory, Upton, New York 11973

(Received 28 August 1987)

We show that certain extended dissipative dynamical systems naturally evolve into a critical state, with no characteristic time or length scales. The temporal "fingerprint" of the self-organized critical state is the presence of flicker noise or $1/f$ noise; its spatial signature is the emergence of scale-invariant (fractal) structure.

Self-Organized Critical State of Sandpile Automaton Models

Deepak Dhar

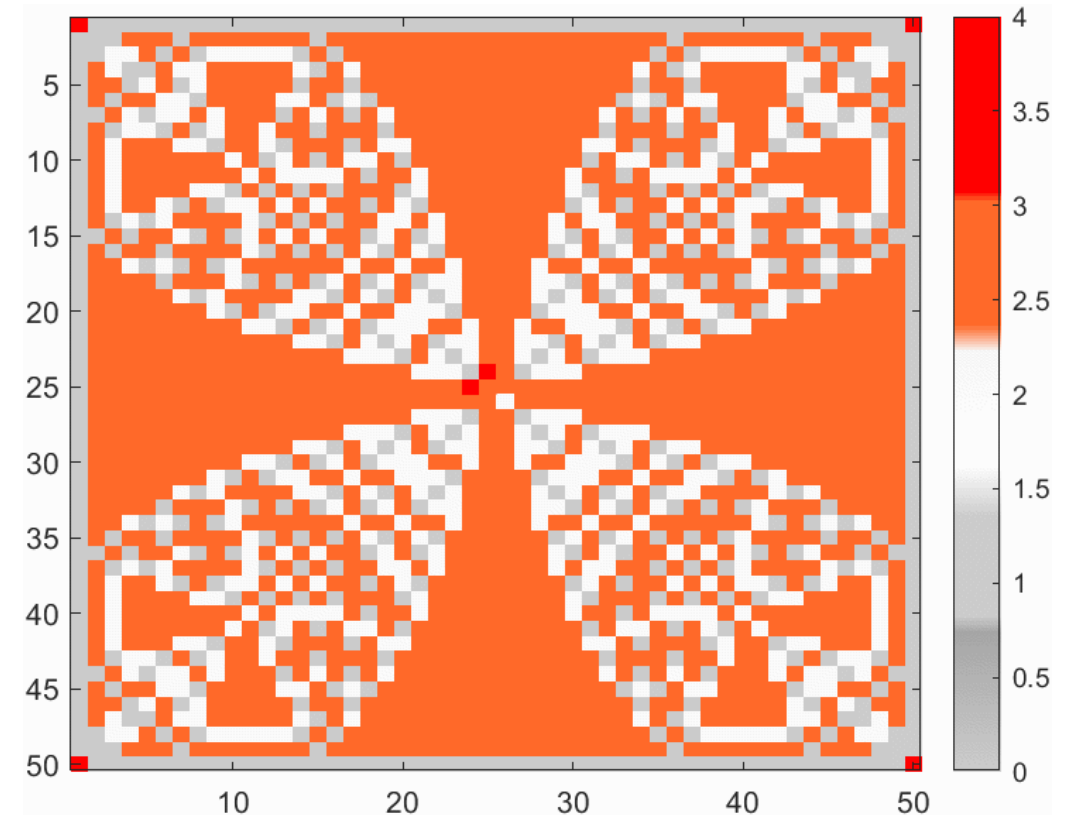
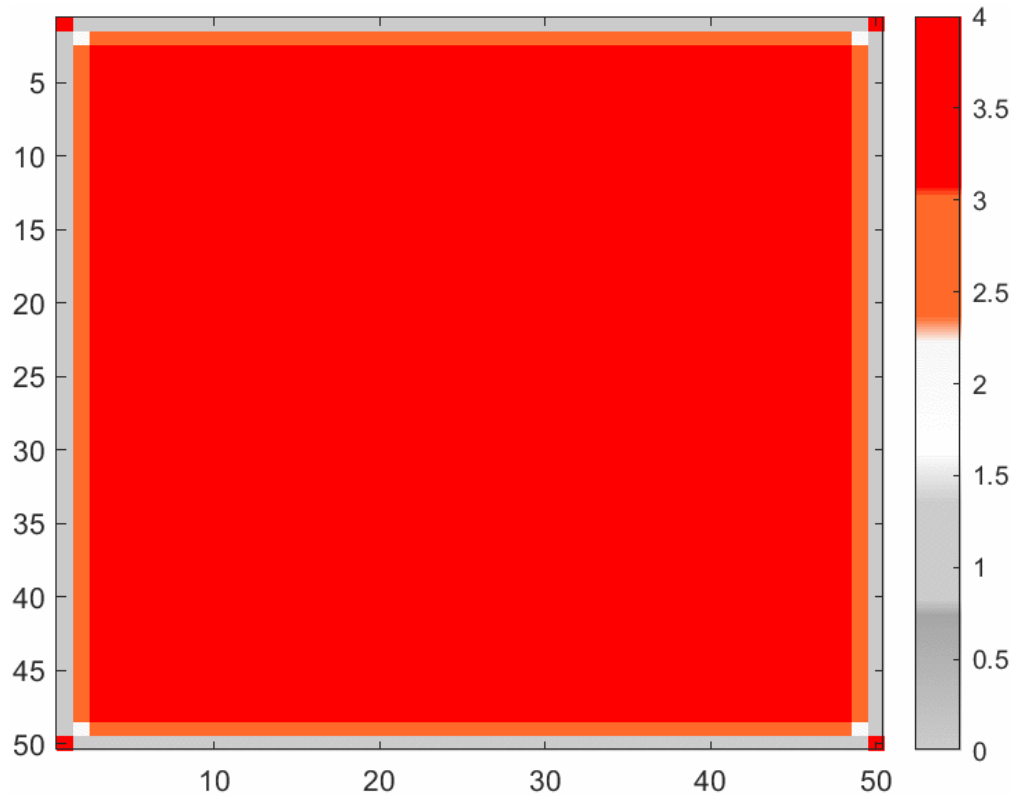
Theoretical Physics Group, Tata Institute of Fundamental Research, Homi Bhabha Road, Bombay 400 005, India

(Received 21 November 1989)

We study a general Bak-Tang-Wiesenfeld-type automaton model of self-organized criticality in which the toppling conditions depend on local height, but not on its gradient. We characterize the critical state, and determine its entropy for an arbitrary finite lattice in any dimension. The two-point correlation function is shown to satisfy a linear equation. The spectrum of relaxation times describing the approach to the critical state is also determined exactly.

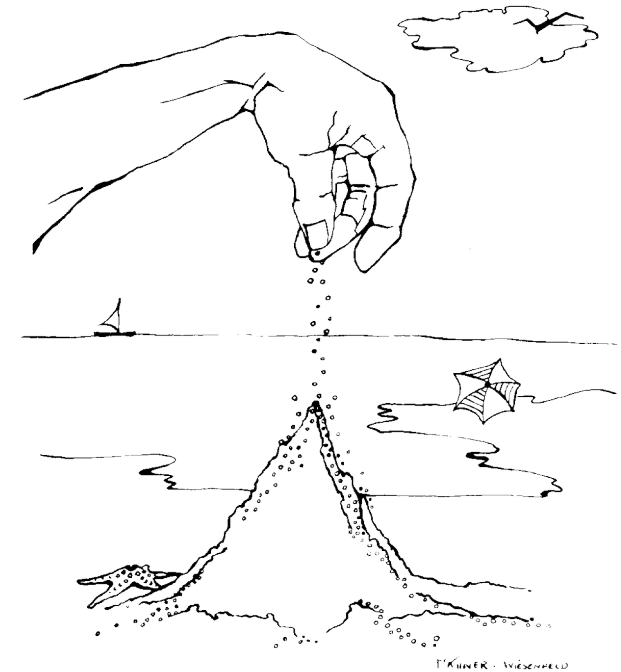
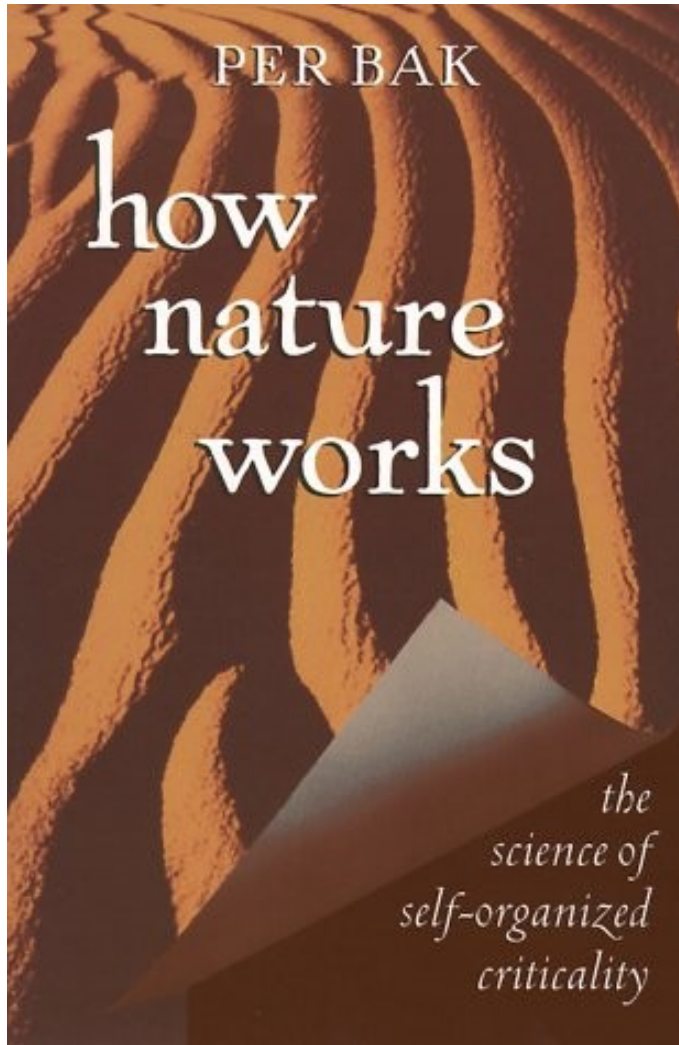
Self-organized criticality

"Who could ever calculate the path of a molecule? How do we know that the creations of worlds are not determined by falling grains of sand?" - Victor Huges, Les Miserables.

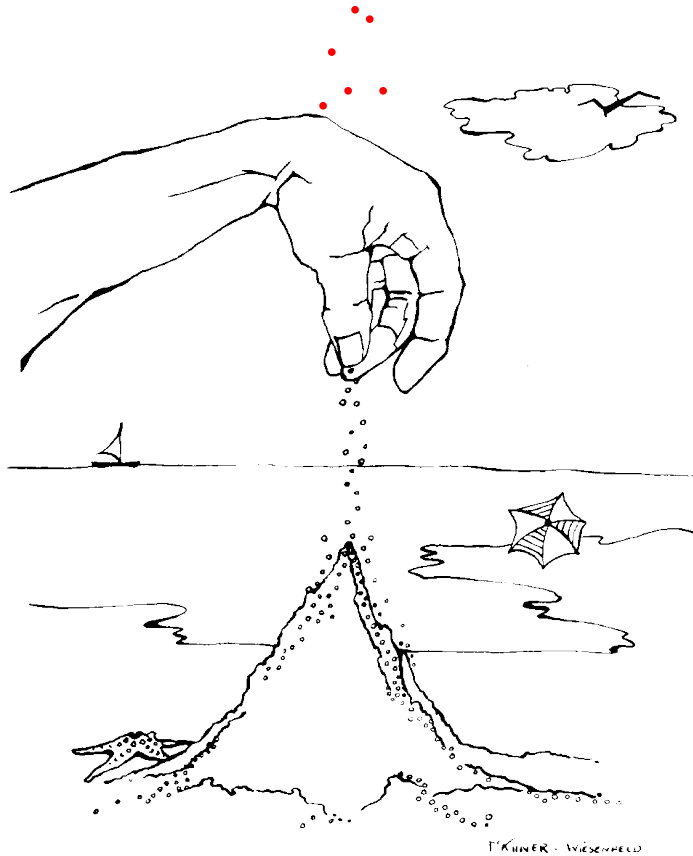


Evolution Via Self Organization

Self Organized Criticality



Avalanche Behavior



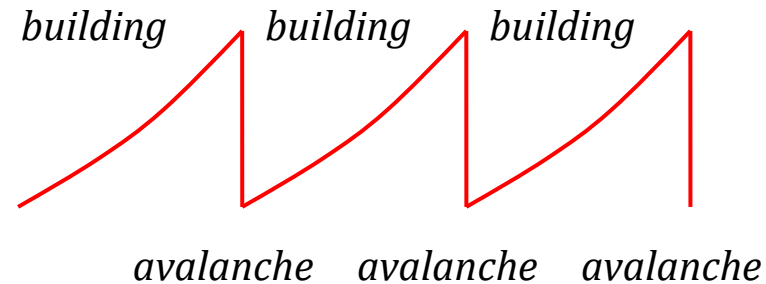
The sand pile builds ... grain ... by grain ...

by grain ... by grain ... by grain ...

by grain ... by grain ... by grain ...

Building toward the critical state ...

Where it avalanches



Avalanche- a large mass of snow, ice, etc., detached from a mountain slope and sliding or falling suddenly downward.

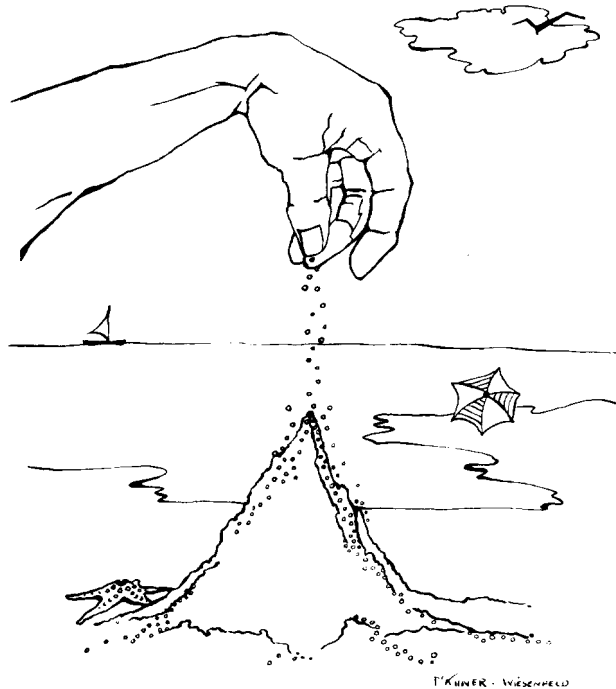
*Avalanche- anything like an **avalanche** in suddenness and overwhelming quantity: an avalanche of misfortunes; an avalanche of fan mail.*

Self-Organized Criticality

Per Bak



1948-2002



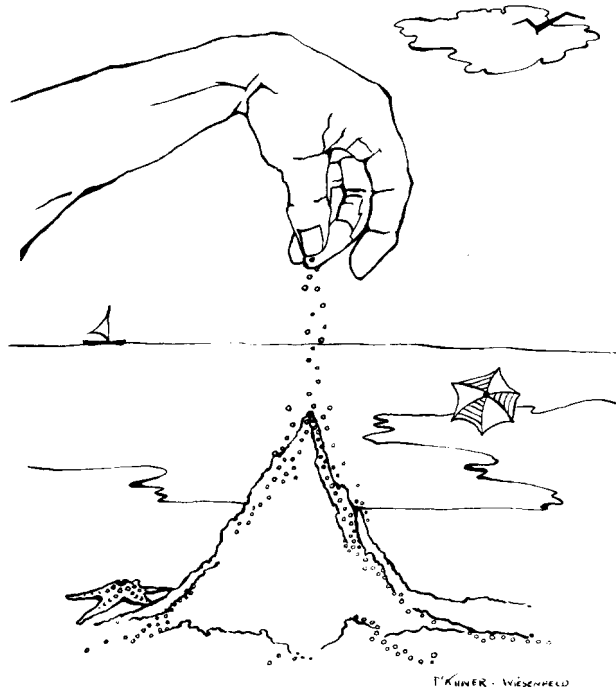
*"Complex behavior in nature reflects the tendency of large systems with many components to evolve into a poised, "critical" state, way out of balance, where minor disturbances may lead to events, called avalanches, **of all sizes**."*

Self-Organized Criticality

Per Bak



1948-2002



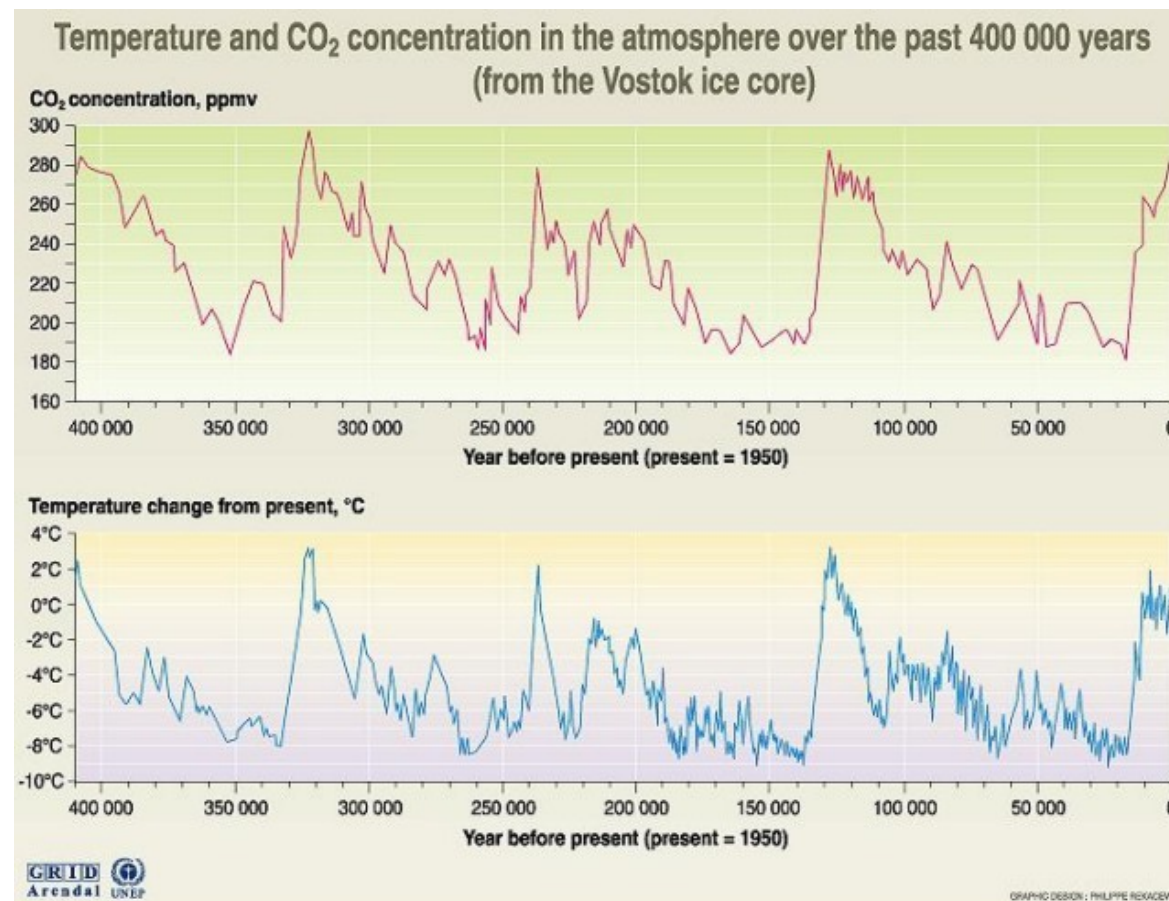
*“Complex behavior in nature reflects the tendency of large systems with many components to evolve into a poised, “critical” state, way out of balance, where minor disturbances may lead to events, called avalanches, **of all sizes**.*

*Most of the changes take place through catastrophic events rather than by following a smooth gradual path. **The evolution to this very delicate state occurs without design from any outside agent.***

*The state is established solely because of the **dynamical interactions among individual elements of the system: the critical state is self-organized. Self-organized criticality is so far the only known general mechanism to generate complexity.***

Now, imagine the sand supply follows a power law (or is fractal), with different numbers of grains falling at different times.

Avalanches will follow a power law distribution.



*Earth Temp. curve
over the past
400,000 years*

Source: J.R. Petit, J. Jouzel, et al. Climate and atmospheric history of the past 420 000 years from the Vostok ice core in Antarctica, Nature 399 (3JUne), pp 429-436, 1999.

Sandpile model

In most systems which exhibit critical phenomena, there exist adjustable parameters which have to be finely tuned in order for the system to reach the critical point. [For example, in the Ising model](#), one needs to adjust both the temperature and the external magnetic field to close to their critical values.

If such a system is initially prepared such that the temperature and the external magnetic field are far from their critical values, the system will not show critical behavior even if other parameters are changed, or if the system is perturbed.

Cellular automata are mathematical models for systems in which simple components act together to form complicated and unpredictable patterns using a set of predefined rules. Two dimensional cellular automata are important to explain pattern formation in many physical systems, including crystal growth, turbulent patterns, and is greatly related to the topic of the sandpile. There are two neighborhoods that are often used in 2 dimensional CA models.

Sandpile model in 2-d Lattice

PHYSICAL REVIEW A

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Self-organized criticality

Per Bak, Chao Tang, and Kurt Wiesenfeld

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(Received 28 August 1987)

We show that certain extended dissipative dynamical systems naturally evolve into a critical state, with no characteristic time or length scales. The temporal “fingerprint” of the self-organized critical state is the presence of flicker noise or $1/f$ noise; its spatial signature is the emergence of scale-invariant (fractal) structure.

Sandpile model in 2-d Lattice

- *Step 1: Allow to make it stable starting from “unstable” states.*
- *Step 2: Role of local perturbation*

6.4 The Sandpile Model and Self-Organized Criticality

181

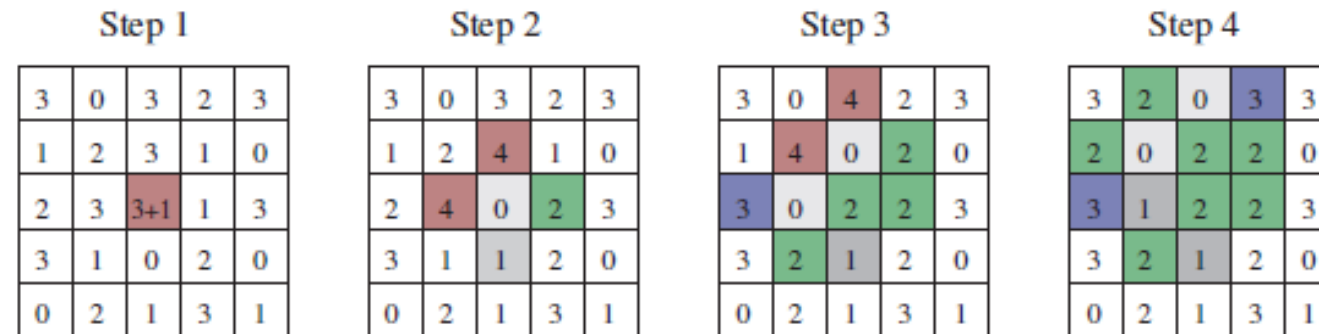


Fig. 6.8 The progress of an avalanche, with duration $t = 3$ and size $s = 13$, for a sandpile configuration on a 5×5 lattice with $K = 3$. The height of the sand in each cell is indicated by the numbers. The shaded region is where the avalanche has progressed. The avalanche stops after step 3

Self-organized criticality

Sandpile model in 2-d Lattice

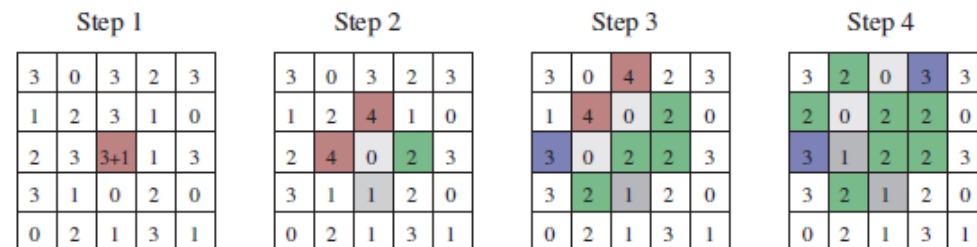


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The Sandpile Model Per Bak and coworkers introduced a simple cellular automaton that mimics the properties of sandpiles, i.e. the BTW model. Every cell is characterized by a force

$$z_i = z(i, j) = 0, 1, 2, \dots, \quad i, j = 1, \dots, L \quad \text{on a finite } L \times L \text{ lattice.}$$

Sandpile model in 2-d Lattice

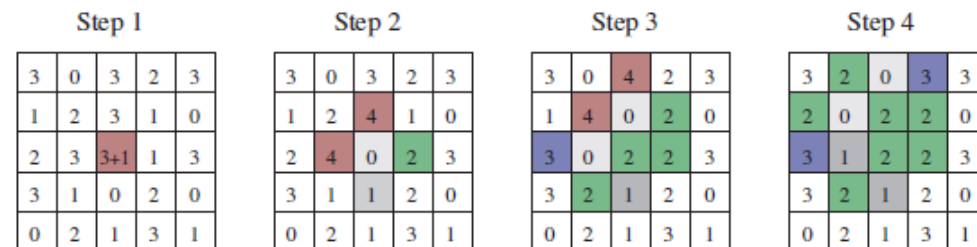


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Loosely speaking one may identify the force z_i with the slope of real-world sandpiles. But this analogy is not rigorous, as the slope of a real-world sandpile is a continuous variable.

Algorithm: Sandpile Avalanche Experiment ($\tau \approx 1$)

Step 1 — Create the lattice

Define a square lattice of size

$$N \times N$$

Each site (i, j) stores a number of sand grains

$$z_{i,j}$$

Initialize the lattice with a large unstable value

$$z_{i,j} = d, d > 3$$

Example

$z(i,j) = 9$, for all sites.

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Example

$z(i,j) = 9$, for all sites.

Step 2 — Stabilize the lattice

Repeatedly apply the toppling rule until every site satisfies

$$z_{i,j} < 4$$

Toppling rule:

$$z_{i,j} \rightarrow z_{i,j} - 4$$

$$z_{i+1,j}, z_{i-1,j}, z_{i,j+1}, z_{i,j-1} \rightarrow +1$$

Grains that leave the boundary disappear (absorbing boundary condition).

This produces a stable sandpile configuration.

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Grains that leave the boundary disappear (absorbing boundary condition).

This produces a **stable sandpile configuration**.

Step 3 — Select a perturbation site

Choose a lattice site

$$(x, y)$$

where the avalanche will start.

Step 4 — Trigger an avalanche

Set

$$z_{x,y} = 4$$

which makes the site **unstable**.

Step 5 — Avalanche propagation

While there exists a site with

$$z_{i,j} \geq 4$$

all unstable sites topple simultaneously (synchronous update).
This produces waves of activity moving outward.

Step 6 — Track avalanche cluster

During the avalanche record all sites that topple.

Define **avalanche size**

$$s = \text{number of distinct sites that topple}$$

Step 7 — Repeat experiment

Repeat Steps 3–6 for many perturbation sites.

Collect avalanche sizes ***s***.

Step 8 — Compute avalanche distribution

Calculate

$$D(s)$$

Plot on logarithmic axes: $D(s)$ vs s using log-binning and log-scale

The distribution approximately follows $D(s) \sim s^{-1}$.

This power-law behavior is characteristic of Self-organized criticality.

Sandpile model in 2-d Numerical model

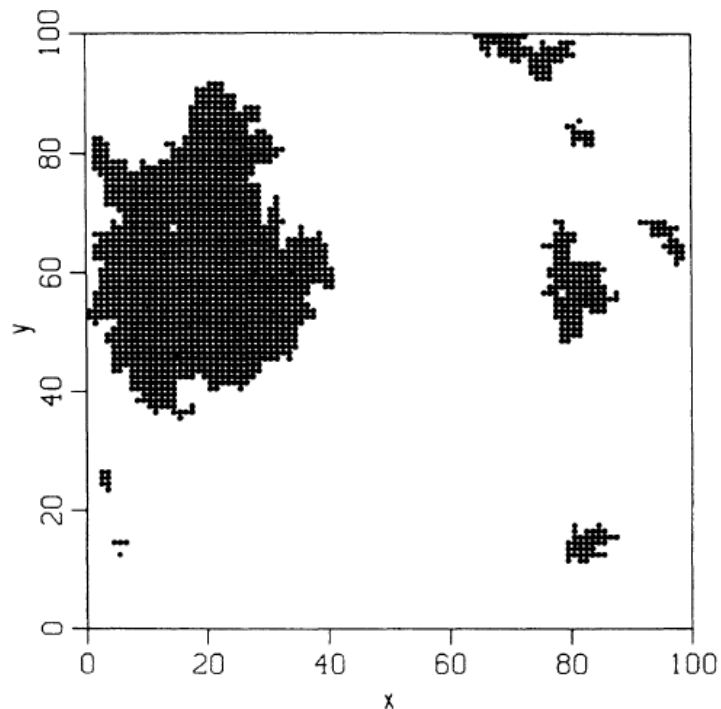


FIG. 2. Typical domain structures resulted from several local perturbations for a 100×100 array. Each cluster is triggered by a single perturbation.

Self-organized criticality

Per Bak, Chao Tang, and Kurt Wiesenfeld

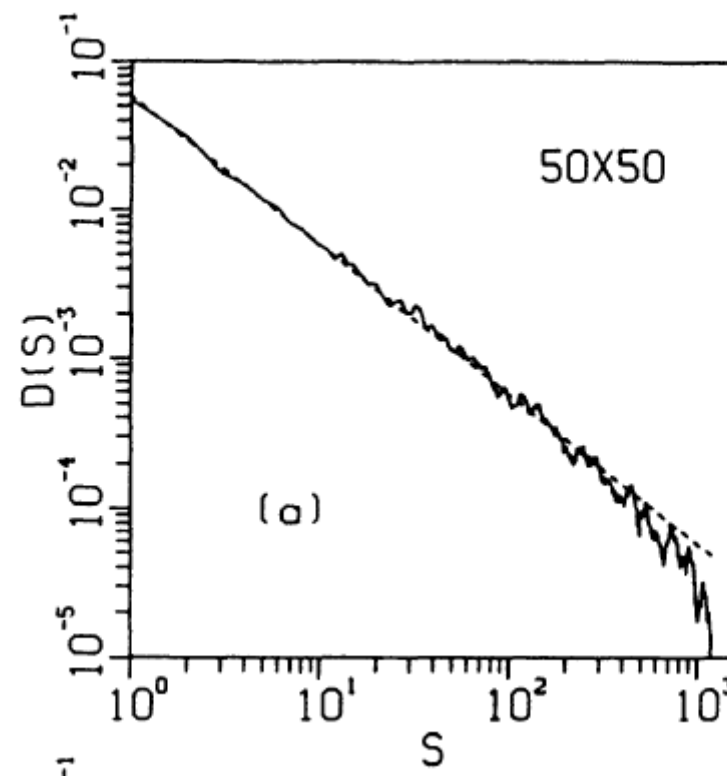


FIG. 3. Distribution of cluster sizes at criticality in two and three dimensions computed as described in the text. The data have been coarse grained. (a) 50×50 array, averaged over 200 samples. The dashed line is a straight line with slope -1.0 ; (b)

Sandpile model in 2-d Lattice

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$$z_i = z(i, j) = 0, 1, 2, \dots, \quad i, j = 1, \dots, L \quad \text{on a finite } L \times L \text{ lattice.}$$

Loosely speaking one may identify the force z_i with the slope of real-world sandpiles. But this analogy is not rigorous, as the slope of a real-world sandpile is a continuous variable.

The slopes belonging to two neighboring cells should therefore be similar, whereas the values of z_i and z_j on two neighboring cells can differ by an arbitrary amount within the sandpile model.

The sand begins to topple when the slope gets too big:

$$z_j \rightarrow z_j - \Delta_{ij}, \quad \text{if } z_i > K ,$$

Sandpile model in 2-d Lattice

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where K is the threshold slope and with the toppling matrix

$$\Delta_{i,j} = \begin{cases} 4 & i = j \\ -1 & i, j \text{ nearest neighbors} \\ 0 & \text{otherwise} \end{cases} .$$

Sandpile model in 2-d Lattice

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$$\Delta_{i,j} = \begin{cases} 4 & i = j \\ -1 & i, j \text{ nearest neighbors} \\ 0 & \text{otherwise} \end{cases} .$$

This update rule is valid for the four-cell neighborhood $\{(0, \pm 1), (\pm 1, 0)\}$. The threshold K is arbitrary, a shift in K simply shifts z_i . It is customary to consider $K = 3$. Any initial random configuration will then relax into a steady-state final configuration (called the stable state) with

$$z_i = 0, 1, 2, 3, \quad (\text{stable state}) .$$

Sandpile model in 2-d Lattice

The sandpile model is locally conserving. The total height $\sum_j z_j$ is constant due to $\sum_j \Delta_{i,j} = 0$. Globally, however, it is not conserving, as one uses open boundary conditions for which excess sand is lost at the boundary. When a site at the boundary topples, some sand is lost there and the total $\sum_j z_j$ is reduced by one.

- **Role of IC:** *Starting from identical stable IC, and perturb in several locations one by one.*

Avalanches When starting from a random initial state with $z_i \ll K$ the system settles in a stable configuration when adding “grains of sand” for a while. When a grain of sand is added to a site with $z_i = K$

$$z_i \rightarrow z_i + 1, \quad z_i = K ,$$

a toppling event is induced, which may in turn lead to a whole series of topplings. The resulting avalanche is characterized by its duration t and the size s of affected sites. It continues until a new stable configuration is reached.

SOC and Power Law

Sandpile model in 2-d Numerical model

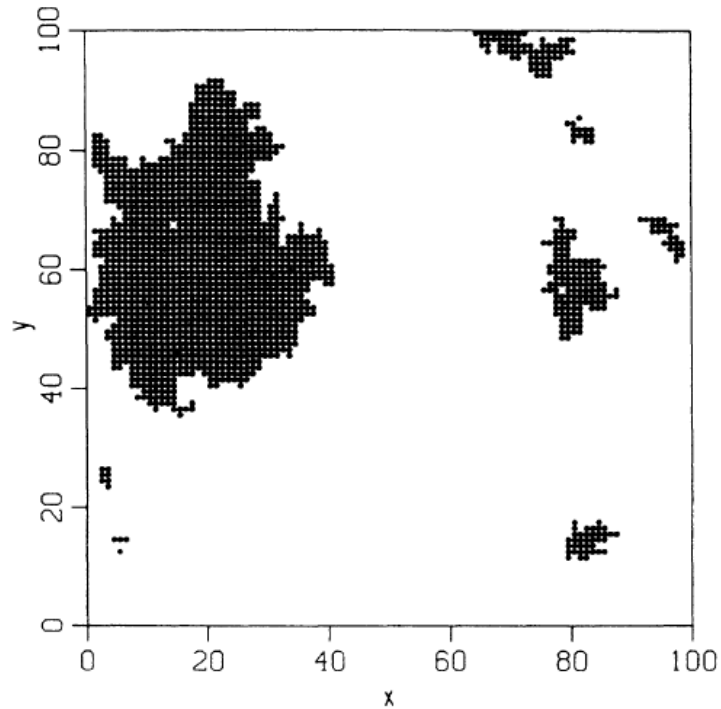


FIG. 2. Typical domain structures resulted from several local perturbations for a 100×100 array. Each cluster is triggered by a single perturbation.

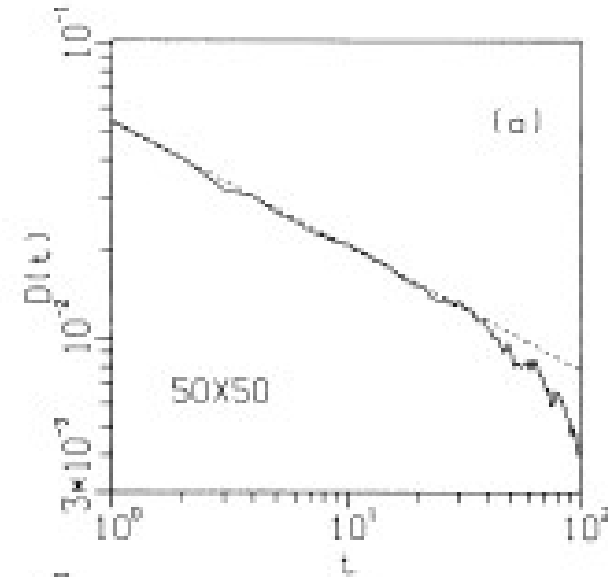


FIG. 3. Distribution of lifetimes (a) For the 50×50 array, the slope $\alpha = 0.42$ ($D(t) = t^{-\alpha}$)

$$D(t) = \frac{s}{t} D(s(t)) \frac{ds}{dt} \approx t^{-(\gamma+1)\tau+2\gamma} \equiv t^{-\alpha}.$$

Self-organized criticality

Per Bak, Chao Tang, and Kurt Wiesenfeld

SOC in Forest-Fire model

Forest Fires: An Example of Self-Organized Critical Behavior

Bruce D. Malamud,* Gleb Morein, Donald L. Turcotte

Despite the many complexities concerning their initiation and propagation, forest fires exhibit power-law frequency-area statistics over many orders of magnitude. A simple forest fire model, which is an example of self-organized criticality, exhibits similar behavior. One practical implication of this result is that the frequency-area distribution of small and medium fires can be used to quantify the risk of large fires, as is routinely done for earthquakes.

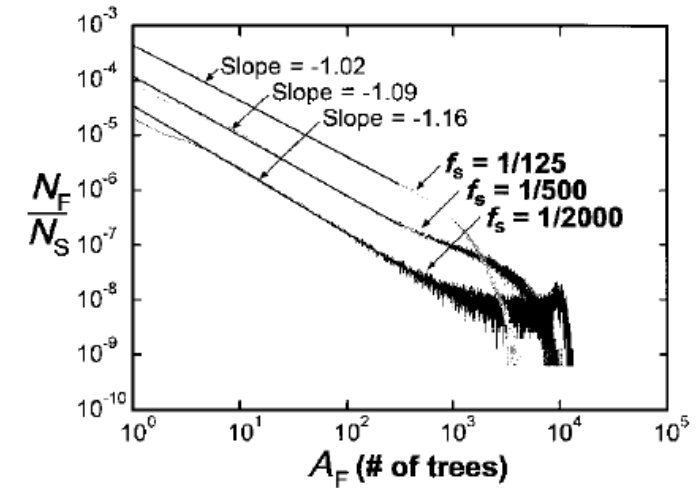


Fig. 1. Noncumulative frequency-area distributions of model forest fires for a grid size of 128 by 128 squares at three sparking frequencies. $f_s = 1/125$, $1/500$, and $1/2000$. The number of fires per time step (N_F/N_S) with area (A_F) is given as a function of A_F , the number of trees that were burned in each fire. For each sparking frequency, the model is run for $N_S = 1.638 \times 10^9$ time steps. The small and medium fires correlate well with the power-law relation (Eq. 1) with $\alpha = 1.02$ to 1.18 ; $-\alpha$ is the slope of the best-fit line in log-log space and is shown for each sparking frequency. The finite grid-size effect can be seen at the smallest sparking frequency, $f_s = 1/2000$. At about $A_F = 2000$, fires begin to span the entire grid.

SOC in Model of Evolution

Punctuated Equilibrium and Criticality in a Simple Model of Evolution

Per Bak

Brookhaven National Laboratory, Upton, New York 11973

Kim Sneppen

Niels Bohr Institute, Blegdamsvej 17, 2100 Copenhagen Ø, Denmark

(Received 7 July 1993)

A simple and robust model of biological evolution of an ecology of interacting species. The model self-organizes into a critical steady state with intermittent coevolution of all sizes; i.e., it exhibits “punctuated equilibrium” behavior. This collaborative evolution is faster than noncooperative scenarios since no large and coordinated, and hence protracted, mutations are involved.

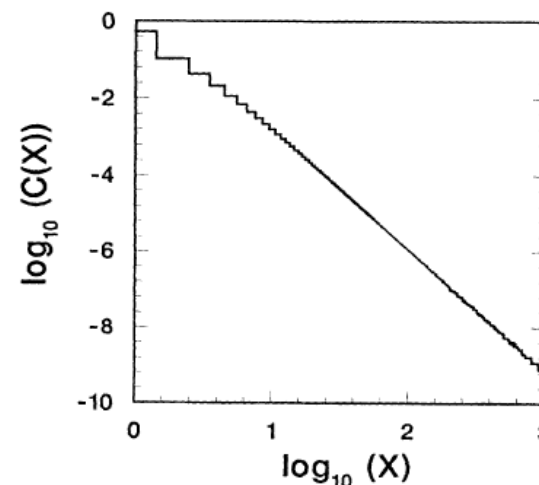


FIG. 1. Distribution of distances $C(x)$ between successive mutations. The power law indicates that the ecology has self-organized into a critical state.

Examples of SOC

RESEARCH ARTICLE

Controlled bio-inspired self-organised criticality

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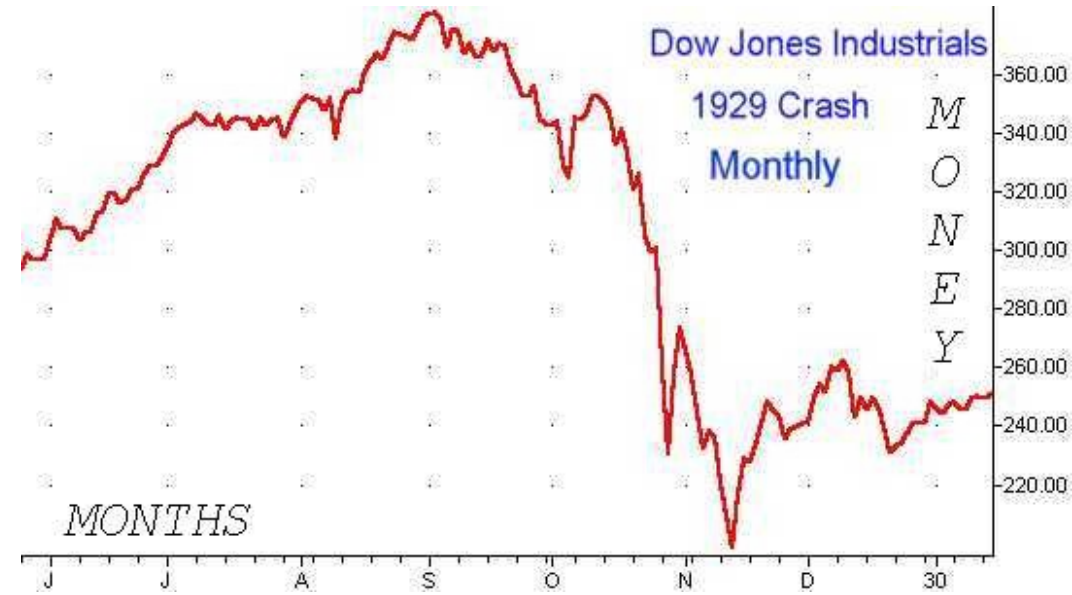
OPEN

A self-organized critical model
and multifractal analysis
for earthquakes in Central Alborz,
Iran

M. Rahimi-Majd¹, T. Shirzad² & M. N. Najafi³⁰²²

*Critical
phenomena*

1929 stock market crash



1987 stock market crash

Examples of Extreme Avalanches

Cascading Power Grids Failures – when a hub is required to carry more than it is capable of carrying, and so crashes, leading to the next hub to crash, etc.

North America blackout 1965

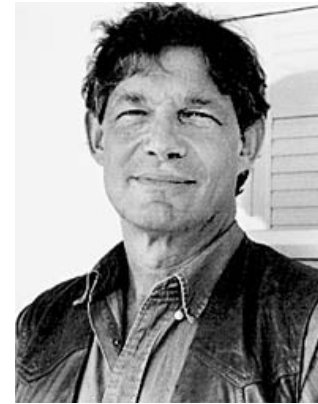


North America blackout 2003



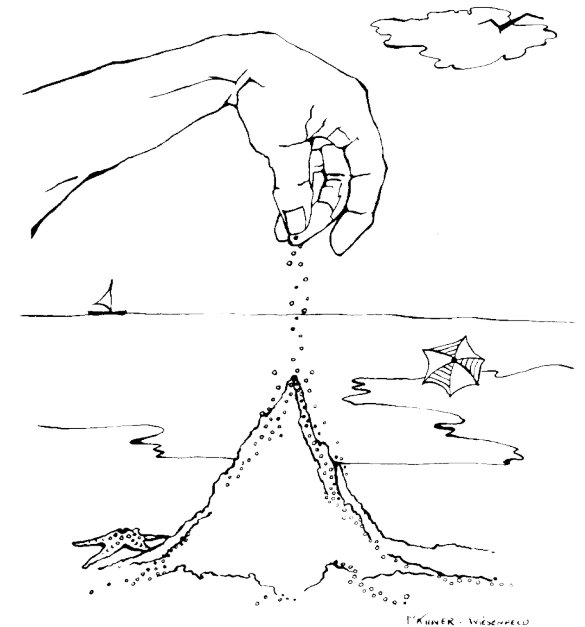
Stuart Kauffman

"The critical point is not, as Stuart Kauffman once described it, "a nice place to be." So "survival of the fittest" does not imply evolution to a state where everybody is well off. On the contrary, individual species are barely able to hang on - like the grains of sand in the critical sand pile."

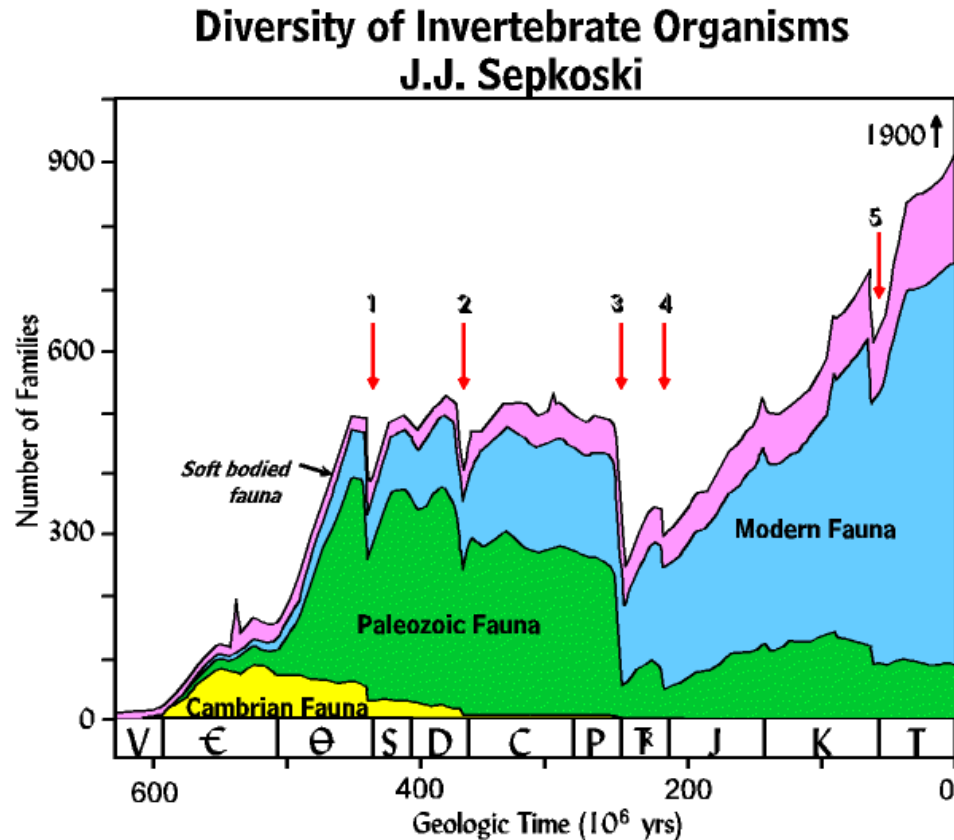


Systems are always at the critical point, or if they are not at the critical point they are evolving toward the critical point.

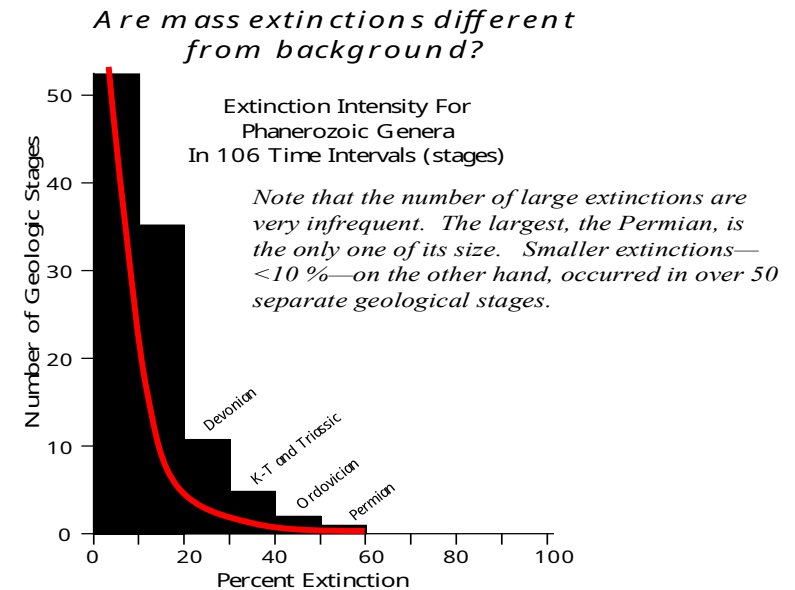
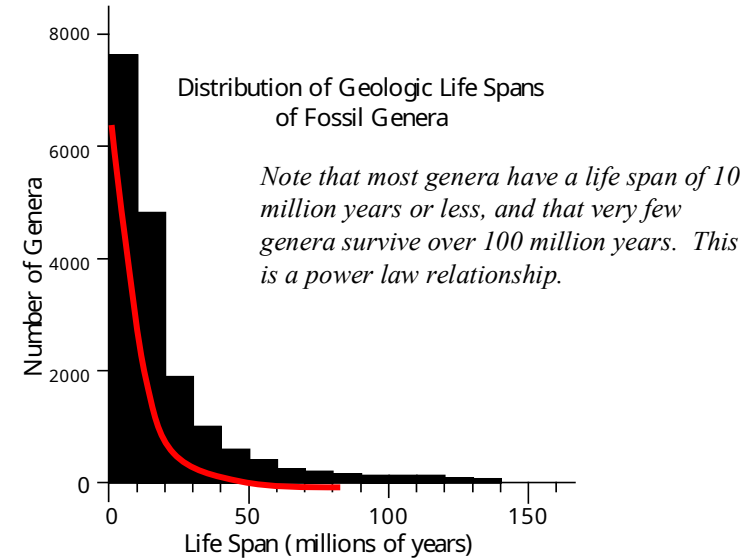
That is, the common idea that systems evolve toward equilibrium is a misperception of reality.



Examples of Extreme Avalanches Extinctions



Life is a Self Organized Critical phenomena



Examples of SOC

RESEARCH ARTICLE

Controlled bio-inspired self-organised criticality

Tjeerd V. olde Scheper  *

School of Engineering, Computing and Mathematics, Oxford Brookes University, Wheatley Campus, Oxford, United Kingdom

*Life is a Self Organized Critical
phenomena*